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Aggregated uplink shared channel transmission for two step random access channel procedure

Abstract

This disclosure relates to methods, devices, and systems for wireless communications, and more particularly to aggregating physical uplink shared channel (PUSCH) resource units relating to a random access message. A user equipment (UE) may identify a random access message of a random access procedure. The random access message may include a random access preamble and a random access payload. The UE may aggregate multiple PUSCH resource units of a set of PUSCH resource units associated with one or more PUSCH occasions (POs) based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload. As a result, the UE may transmit the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs.

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Background/Summary

CROSS-REFERENCE TO REALTED APPLICATIONS
(1) The present patent Application is a 371 national stage filing of International PCT Application No. PCT/CN2020/087429 by Cao et al., entitled “AGGREGATED UPLINK SHARED CHANNEL TRANSMISSION FOR TWO STEP RANDOM ACCESS CHANNEL PROCEDURE,” filed Apr. 28, 2020; and claims priority to International PCT Application No. PCT/CN2019/085436 by Cao et al., entitled “AGGREGATED UPLINK SHARED CHANNEL TRANSMISSION FOR TWO STEP RANDOM ACCESS CHANNEL PROCEDURE,” filed May 4, 2019, each of which is assigned to the assignee hereof, and each of which is expressly incorporated by reference in its entirety herein.
BACKGROUND
(2) The following relates generally to wireless communications, and more specifically to aggregating multiple physical uplink shared channel (PUSCH) resource units relating to a random access message of a random access procedure.
(3) Wireless communications systems are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be capable of supporting communication with multiple users by sharing the available system resources (e.g., time, frequency, and power). Examples of such multiple-access systems include fourth generation (4G) systems such as Long Term Evolution (LTE) systems, LTE-

Advanced (LTE-A) systems, or LTE-A Pro systems, and fifth generation (5G) systems which may be referred to as New Radio (NR) systems. These systems may employ technologies such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal frequency division multiple access (OFDMA), or discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-S-OFDM).

SUMMARY

(4) The described techniques relate to improved methods, systems, devices, and apparatuses that support techniques related to a random access procedure (e.g., a two-step random access procedure), and more particularly to aggregating physical uplink shared channel (PUSCH) resource units relating to a random access message in the random access procedure.

(5) A wireless multiple-access communications system may include a number of base stations or network access nodes, each supporting communication for multiple communication devices, which may be otherwise known as user equipments (UE). Some wireless communications systems may support one or more random access procedures, for example, for communication between a UE and a base station. The random access procedures may involve a series of messages exchanged between the UE and the base station.

(6) Generally, the described techniques support aggregating multiple PUSCH resources units to support larger payload sizes of random access messages or reduce a modulation and coding scheme of the random access messages. In some examples, the described techniques may include identifying a random access message of a random access procedure. The random access message may include a random access preamble and a random access payload. The described techniques may include aggregating multiple PUSCH resource units associated with one or more physical uplink shared channel occasions (POs) based on one or more of a payload size of the random access payload, or a modulation and coding scheme associated with the random access payload, or other factors or conditions.

(7) Additionally, the described techniques may include, in some examples, transmitting the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs. Additionally or alternatively, the described techniques may include transmitting the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs using a same transmit power or a same modulation and coding scheme. The described techniques may therefore include features for improved resource usage and allocation for one or more random access messages, improved reliability for random access messaging and, in some examples, may promote low latency for random access procedures, among other benefits.

(8) A method of wireless communication at a user equipment (UE) is described. The method may include identifying a random access message of a random access procedure, the random access message including a random access preamble and a random access payload, aggregating multiple PUSCH resource units of a set of PUSCH resource units associated with one or more POs based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload, and transmitting the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs.

(9) An apparatus for wireless communication is described. The apparatus may include a processor, memory in electronic communication with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to identify a random access message of a random access procedure, the random access message including a random access preamble and a random access payload, aggregate multiple PUSCH resource units of a set of PUSCH resource units associated with one or more POs based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access

payload, and transmit the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs.
(10) Another apparatus for wireless communication is described. The apparatus may include means for identifying a random access message of a random access procedure, the random access message including a random access preamble and a random access payload, aggregating multiple PUSCH resource units of a set of PUSCH resource units associated with one or more POs based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload, and transmitting the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs.
(11) A non-transitory computer-readable medium storing code for wireless communication at a UE is described. The code may include instructions executable by a processor to identify a random access message of a random access procedure, the random access message including a random access preamble and a random access payload, aggregate multiple PUSCH resource units of a set of PUSCH resource units associated with one or more POs based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload, and transmit the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs.
(12) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from a base station, an indication of an aggregation configuration for the PUSCH resource units, and where aggregating the multiple PUSCH resource units of the set of PUSCH resource units may be further based on the aggregation configuration.
(13) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, receiving the indication of the aggregation configuration for the PUSCH resource units may include operations, features, means, or instructions for receiving radio resource control (RRC) signaling.
(14) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, receiving the indication may include operations, features, means, or instructions for receiving one or more of a random access response (RAR) window for a random access response associated with the random access procedure or a retransmission period of the random access message.
(15) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from a base station, an indication of one or more of a random access channel occasion (RO) configuration or a PO configuration, and mapping one or more ROs to the one or more POs based on the one or more of RO configuration or the PO configuration.
(16) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, one or more of the one or more ROs or the one or more POs may be contiguous in one or more of the time domain or the frequency domain.
(17) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, one or more of the one or more ROs or the one or more POs may be noncontiguous in one or more of the time domain or the frequency domain.
(18) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, mapping the one or more ROs to the one or more POs may include operations, features, means, or instructions for mapping the random access preamble to the one or more POs in one or more of the time domain or the frequency domain based on the RO configuration or the PO configuration.

(19) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, mapping the one or more ROs to the one or more POs may include operations, features, means, or instructions for assigning available random access preambles including the random access preamble of the random access message associated with one or more ROs to the one or more POs, and mapping the assigned random access preambles to the one or more POs.
(20) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the RO configuration may include operations, features, means, or instructions for a periodicity of an RO, a time and frequency resource allocation of the RO, a preamble sequence configuration for a contention-based random access (CBRA) procedure or a contention-free random access (CFRA) procedure, an association rule between ROs and POs, or a beam association rule between a synchronization signal blocks (SSBs) or a channel state information reference signal (CSI-RS) and the RO.
(21) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the PO configuration may include operations, features, means, or instructions for a periodicity of a PO, a time and frequency resource allocation of the PO or a PUSCH resource unit, a demodulation reference signal (DMRS) configuration, a PUSCH waveform configuration, beam management information for the PUSCH.
(22) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for mapping the random access payload to one or more of the one or more ROs or the one or more POs in one or more of the time domain or the frequency domain based on a hopping sequence.
(23) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, transmitting the random access payload of the random access message on the PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs may include operations, features, means, or instructions for transmitting the random access payload of the random access message on time and frequency resources of the aggregated multiple PUSCH resource units using a same modulation and coding scheme for each PUSCH resource unit of the aggregated multiple PUSCH resource units.
(24) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining a modulation and coding scheme capability for the random access message including the random access preamble and the random access payload, and where aggregating the multiple PUSCH resource units of the set of PUSCH resource units associated with the one or more POs may be further based on the modulation and coding scheme capability.
(25) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, a payload size of one or more PUSCH resource units of the set of PUSCH resource units may be fixed based on the modulation and coding scheme capability supporting a single modulation and coding scheme.
(26) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, a payload size of one or more PUSCH resource units of the set of PUSCH resource units may be variable based on the modulation and coding scheme capability supporting multiple modulation and coding schemes.
(27) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, transmitting the random access payload of the random access message on the PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs may include operations, features, means, or instructions for transmitting, in one or more of the random access preamble or a control portion of the random access payload, an indication of the time and frequency resources of the aggregated multiple PUSCH resource units based on the modulation and coding scheme capability supporting the multiple modulation and coding schemes.

(28) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for mapping a set of DMRSs to random access preambles of PUSCH resource units of the aggregated multiple PUSCH resource units based on a mapping rule.
(29) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, each PUSCH resource unit of the aggregated multiple PUSCH resource units may be associated with a unique DMRS of the set of DMRSs.
(30) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the aggregated multiple PUSCH resource units share a same DMRS of the set of DMRSs.
(31) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, aggregating the multiple PUSCH resource units of the set of PUSCH resource units associated with the one or more POs may include operations, features, means, or instructions for comparing the payload size of the random access payload to an aggregated payload size of the aggregated multiple PUSCH resource units, determining a quantity of unused bits of the aggregated payload size of the aggregated multiple PUSCH resource units based on the comparing, and padding each unused bit of the quantity of unused bits of the aggregated payload size of the aggregated multiple PUSCH resource units with a null bit.
(32) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from a base station, an indication of a power configuration, where transmitting the random access payload of the random access message on the PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs includes transmitting the random access payload of the random access message on time and frequency resources of the aggregated multiple PUSCH resource units using a same transmit power for each PUSCH resource unit of the aggregated multiple PUSCH resource units.
(33) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, one or more PUSCH resource units of the set of PUSCH resource units may have a default payload size.
(34) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the default payload size includes one or more bits.
(35) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, transmitting the random access payload of the random access message on the PUSCH using time and frequency resources may include operations, features, means, or instructions for transmitting using the time and frequency resources that may be consecutive or nonconsecutive in one or more of a time domain or a frequency domain.
(36) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the random access procedure includes a two-step random access procedure.
(37) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving a random access response message including an indication to perform a random access fallback procedure, where the random access fallback procedure includes a four-step random access procedure, and performing the random access fallback procedure based on receiving the random access response message.
(38) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving a random access response message including an indication to retransmit the random access message, and retransmitting the random access message based on the random access response message.
(39) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, a PUSCH resource unit of the set of PUSCH resource units spans a PO.

(40) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the aggregated multiple PUSCH resource units span over multiple POs.
(41) A method of wireless communication at a base station is described. The method may include assigning a set of PUSCH resource units associated with one or more POs for transmission of a random access payload of a random access message associated with a random access procedure, determining an aggregation configuration for multiple PUSCH resource units of the set of PUSCH resource units based on assigning the set of PUSCH resource units, and transmitting, to a UE, an indication of the aggregation configuration for the multiple PUSCH resource units.
(42) An apparatus for wireless communication is described. The apparatus may include a processor, memory in electronic communication with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to assign a set of PUSCH resource units associated with one or more POs for transmission of a random access payload of a random access message associated with a random access procedure, determine an aggregation configuration for multiple PUSCH resource units of the set of PUSCH resource units based on assigning the set of PUSCH resource units, and transmit, to a UE, an indication of the aggregation configuration for the multiple PUSCH resource units.
(43) Another apparatus for wireless communication is described. The apparatus may include means for assigning a set of PUSCH resource units associated with one or more POs for transmission of a random access payload of a random access message associated with a random access procedure, determining an aggregation configuration for multiple PUSCH resource units of the set of PUSCH resource units based on assigning the set of PUSCH resource units, and transmitting, to a UE, an indication of the aggregation configuration for the multiple PUSCH resource units.
(44) A non-transitory computer-readable medium storing code for wireless communication at a base station is described. The code may include instructions executable by a processor to assign a set of PUSCH resource units associated with one or more POs for transmission of a random access payload of a random access message associated with a random access procedure, determine an aggregation configuration for multiple PUSCH resource units of the set of PUSCH resource units based on assigning the set of PUSCH resource units, and transmit, to a UE, an indication of the aggregation configuration for the multiple PUSCH resource units.
(45) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from the UE, a modulation and coding scheme capability for the random access message including the random access preamble and the random access payload, and where determining the aggregation configuration for the multiple PUSCH resource units of the set of PUSCH resource units may be further based on the modulation and coding scheme capability.
(46) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, transmitting the indication of the aggregation configuration for the multiple PUSCH resource units may include operations, features, means, or instructions for transmitting RRC signaling.
(47) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, transmitting the indication may include operations, features, means, or instructions for transmitting one or more of a RAR window for a random access response associated with the random access procedure or a retransmission period of the random access message.
(48) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting, to the UE, a second indication of one or more of an RO configuration or a PO configuration, and mapping one or more ROs to the one or more POs based on the one or more of RO configuration or the PO configuration.
(49) In some examples of the method, apparatuses, and non-transitory computer-readable medium

described herein, one or more of the one or more ROs or the one or more POs may be noncontiguous in one or more of the time domain or the frequency domain.
(50) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, one or more of the one or more ROs or the one or more POs may be contiguous in one or more of the time domain or the frequency domain.
(51) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, mapping the one or more ROs to the one or more POs may include operations, features, means, or instructions for mapping the random access preamble to the one or more POs in one or more of the time domain or the frequency domain based on the RO configuration or the PO configuration.
(52) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the RO configuration may include operations, features, means, or instructions for a periodicity of an RO, a time and frequency resource allocation of the RO, a preamble sequence configuration for a CBRA procedure or a CFRA procedure, an association rule between ROs and POs, or a beam association rule between a SSBs or a CSI-RS and the RO.
(53) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the PO configuration may include operations, features, means, or instructions for a periodicity of a PO, a time and frequency resource allocation of the PO or a PUSCH resource unit, a DMRS configuration, a PUSCH waveform configuration, beam management information for a PUSCH.
(54) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from the UE, the random access payload of the random access message on a PUSCH using time and frequency resources of aggregated multiple PUSCH resource units over one or more POs.
(55) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting, to the UE, a second indication of a power configuration, where receiving the random access payload of the random access message on the PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs includes receiving the random access payload of the random access message on time and frequency resources of the aggregated multiple PUSCH resource units using a same power configuration for each PUSCH resource unit of the aggregated multiple PUSCH resource units.
(56) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, one or more PUSCH resource units of the set of PUSCH resource units may have a default payload size.
(57) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the default payload size includes one or more bits.
(58) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the random access procedure includes a two-step random access procedure.
(59) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting a random access response message including an indication to perform a random access fallback procedure, where the random access fallback procedure includes a four-step random access procedure, and performing the random access fallback procedure based on the random access response message.
(60) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting a random access response message including an indication to the UE to retransmit the random access message.
(61) In some examples of the method, apparatuses, and non-transitory computer-readable medium

described herein, a PUSCH resource unit of the set of PUSCH resource units spans a PO.
(62) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the aggregated multiple physical uplink shared channel resource units span over multiple POs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** illustrates an example of a wireless communications system that supports aggregated uplink shared channel transmission for two-step random access channel procedures in accordance with one or more aspects of the present disclosure.
(2) FIGS. **2**A and **2**B illustrate examples of wireless communications systems that support aggregated uplink shared channel transmission for two-step random access channel procedures in accordance with one or more aspects of the present disclosure.
(3) FIG. **3** illustrates an example of a process flow that supports aggregated uplink shared channel transmission for two-step random access channel procedures in accordance with one or more aspects of the present disclosure.
(4) FIGS. **4** and **5** show block diagrams of devices that support aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure.
(5) FIG. **6** shows a block diagram of a UE communications manager that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure.
(6) FIG. **7** shows a diagram of a system including a device that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure.
(7) FIGS. **8** and **9** show block diagrams of devices that support aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure.
(8) FIG. **10** shows a block diagram of a base station communications manager that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure.
(9) FIG. **11** shows a diagram of a system including a device that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure.
(10) FIGS. **12** through **15** show flowcharts illustrating methods that support aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure.
DETAILED DESCRIPTION
(11) Some wireless communications systems may have one or more user equipments (UEs) and one or more base stations, for example, next-generation NodeBs or giga-NodeBs (either of which may be referred to as a gNB) that support one or more random access procedures for communication, including an initial access to a channel, a connection re-establishment, a handover procedure, or synchronization on the channel, among others. The random access procedure may include a series of handshake messages, such as random access messages carrying information that may facilitate the communication between the UE and the base station. In some examples, a random access procedure may be or may include a two-step random access procedure, which may reduce latency compared to other random access procedures that use a greater number of handshake messages, for example, such as a four-step random access procedure. As demand for

communication efficiency increases, it may be desirable for a UE to seek improved reliability for random access messaging, as well as target efficient resource usage for random access procedures, particularly for variable payload sizes of different random access messages.
(12) As described herein, UEs and base stations may support aggregating multiple PUSCH resources units to support larger payload sizes of random access messages or reduce a modulation and coding scheme of the random access messages, among other benefits. In some examples, a UE may identify a random access message of a random access procedure. The random access message may include a random access preamble and a random access payload. The UE may aggregate multiple physical uplink shared channel (PUSCH) resource units associated with one or more physical uplink shared channel occasions (POs) based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload, or other factors or conditions. Additionally, the UE may transmit the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs.
(13) Particular aspects of the subject matter described herein may be implemented to realize one or more advantages. The described techniques may support improvements in resource usage (e.g., time and frequency resources) for random access messages, among other advantages. As such, supported techniques may include features for efficient random access procedures. The described techniques may also support improved reliability for random access messaging and, in some examples, may promote low latency for random access procedures, among other benefits.
(14) Aspects of the disclosure are initially described in the context of a wireless communications system. Aspects of the disclosure are further described in the context of one or more additional wireless communications systems and one or more process flows that relate to aspects for aggregated uplink shared channel transmission for two-step random access channel procedures. Aspects of the disclosure are further illustrated by and described with reference to apparatus diagrams, system diagrams, and flowcharts that relate to several aspects related to aggregated uplink shared channel transmission for two-step random access channel procedures.
(15) FIG. **1** illustrates an example of a wireless communications system **100** that supports aggregated uplink shared channel transmission for two-step random access channel procedures in accordance with one or more aspects of the present disclosure. The wireless communications system **100** includes base stations **105**, UEs **115**, and a core network **130**. In some examples, the wireless communications system **100** may be a Long Term Evolution (LTE) network, an LTE-Advanced (LTE-A) network, an LTE-A Pro network, or a New Radio (NR) network. In some cases, wireless communications system **100** may support enhanced broadband communications, ultra-reliable (e.g., mission critical) communications, low latency communications, or communications with low-cost and low-complexity devices.
(16) Base stations **105** may wirelessly communicate with UEs **115** via one or more base station antennas. Base stations **105** described herein may include or may be referred to by those skilled in the art as a base transceiver station, a radio base station, an access point, a radio transceiver, a NodeB, an eNodeB (eNB), a next-generation NodeB or giga-NodeB (either of which may be referred to as a gNB), a Home NodeB, a Home eNodeB, or some other suitable terminology. Wireless communications system **100** may include base stations **105** of different types (e.g., macro or small cell base stations). The UEs **115** described herein may be able to communicate with various types of base stations **105** and network equipment including macro eNBs, small cell eNBs, gNBs, relay base stations, and the like.
(17) Each base station **105** may be associated with a particular geographic coverage area **110** in which communications with various UEs **115** is supported. Each base station **105** may provide communication coverage for a respective geographic coverage area **110** via communication links **125**, and communication links **125** between a base station **105** and a UE **115** may utilize one or more carriers. Communication links **125** shown in wireless communications system **100** may

include uplink transmissions from a UE **115** to a base station **105**, or downlink transmissions from a base station **105** to a UE **115**. Downlink transmissions may also be called forward link transmissions while uplink transmissions may also be called reverse link transmissions.

(18) The geographic coverage area **110** for a base station **105** may be divided into sectors making up a portion of the geographic coverage area **110**, and each sector may be associated with a cell. For example, each base station **105** may provide communication coverage for a macro cell, a small cell, a hot spot, or other types of cells, or various combinations thereof. In some examples, a base station **105** may be movable and therefore provide communication coverage for a moving geographic coverage area **110**. In some examples, different geographic coverage areas **110** associated with different technologies may overlap, and overlapping geographic coverage areas **110** associated with different technologies may be supported by the same base station **105** or by different base stations **105**. The wireless communications system **100** may include, for example, a heterogeneous LTE/LTE-A/LTE-A Pro or NR network in which different types of base stations **105** provide coverage for various geographic coverage areas **110**.

(19) The term “cell” refers to a logical communication entity used for communication with a base station **105** (e.g., over a carrier), and may be associated with an identifier for distinguishing neighboring cells (e.g., a physical cell identifier (PCID), a virtual cell identifier (VCID)) operating via the same or a different carrier. In some examples, a carrier may support multiple cells, and different cells may be configured according to different protocol types (e.g., machine-type communication (MTC), narrowband Internet-of-Things (NB-IoT), enhanced mobile broadband (eMBB), or others) that may provide access for different types of devices. In some cases, the term “cell” may refer to a portion of a geographic coverage area **110** (e.g., a sector) over which the logical entity operates.

(20) UEs **115** may be dispersed throughout the wireless communications system **100**, and each UE **115** may be stationary or mobile. A UE **115** may also be referred to as a mobile device, a wireless device, a remote device, a handheld device, or a subscriber device, or some other suitable terminology, where the “device” may also be referred to as a unit, a station, a terminal, or a client. A UE **115** may also be a personal electronic device such as a cellular phone, a personal digital assistant (PDA), a tablet computer, a laptop computer, or a personal computer. In some examples, a UE **115** may also refer to a wireless local loop (WLL) station, an Internet of Things (IoT) device, an Internet of Everything (IoE) device, or an MTC device, or the like, which may be implemented in various articles such as appliances, vehicles, meters, or the like.

(21) Some UEs **115**, such as MTC or IoT devices, may be low cost or low complexity devices, and may provide for automated communication between machines (e.g., via Machine-to-Machine (M2M) communication). M2M communication or MTC may refer to data communication technologies that allow devices to communicate with one another or a base station **105** without human intervention. In some examples, M2M communication or MTC may include communications from devices that integrate sensors or meters to measure or capture information and relay that information to a central server or application program that can make use of the information or present the information to humans interacting with the program or application. Some UEs **115** may be designed to collect information or enable automated behavior of machines. Examples of applications for MTC devices include smart metering, inventory monitoring, water level monitoring, equipment monitoring, healthcare monitoring, wildlife monitoring, weather and geological event monitoring, fleet management and tracking, remote security sensing, physical access control, and transaction-based business charging.

(22) Some UEs **115** may be configured to employ operating modes that reduce power consumption, such as half-duplex communications (e.g., a mode that supports one-way communication via transmission or reception, but not transmission and reception simultaneously). In some examples, half-duplex communications may be performed at a reduced peak rate. Other power conservation techniques for UEs **115** include entering a power saving “deep sleep” mode when not engaging in

active communications, or operating over a limited bandwidth (e.g., according to narrowband communications). In some cases, UEs **115** may be designed to support critical functions (e.g., mission critical functions), and a wireless communications system **100** may be configured to provide ultra-reliable communications for these functions.
(23) In some cases, a UE **115** may also be able to communicate directly with other UEs **115** (e.g., using a peer-to-peer (P2P) or device-to-device (D2D) protocol). One or more of a group of UEs **115** utilizing D2D communications may be within the geographic coverage area **110** of a base station **105**. Other UEs **115** in such a group may be outside the geographic coverage area **110** of a base station **105**, or be otherwise unable to receive transmissions from a base station **105**. In some cases, groups of UEs **115** communicating via D2D communications may utilize a one-to-many (1:M) system in which each UE **115** transmits to every other UE **115** in the group. In some cases, a base station **105** facilitates the scheduling of resources for D2D communications. In other cases, D2D communications are carried out between UEs **115** without the involvement of a base station **105**.
(24) Base stations **105** may communicate with the core network **130** and with one another. For example, base stations **105** may interface with the core network **130** through backhaul links **132** (e.g., via an S1, N2, N3, or other interface). Base stations **105** may communicate with one another over backhaul links **134** (e.g., via an X2, Xn, or other interface) either directly (e.g., directly between base stations **105**) or indirectly (e.g., via core network **130**).
(25) The core network **130** may provide user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. The core network **130** may be an evolved packet core (EPC), which may include at least one mobility management entity (MME), at least one serving gateway (S-GW), and at least one Packet Data Network (PDN) gateway (P-GW). The MME may manage non-access stratum (e.g., control plane) functions such as mobility, authentication, and bearer management for UEs **115** served by base stations **105** associated with the EPC. User IP packets may be transferred through the S-GW, which itself may be connected to the P-GW. The P-GW may provide IP address allocation as well as other functions. The P-GW may be connected to the network operators IP services. The operators IP services may include access to the Internet, Intranet(s), an IP Multimedia Subsystem (IMS), or a Packet-Switched (PS) Streaming Service.
(26) At least some of the network devices, such as a base station **105**, may include subcomponents such as an access network entity, which may be an example of an access node controller (ANC). Each access network entity may communicate with UEs **115** through a number of other access network transmission entities, which may be referred to as a radio head, a smart radio head, or a transmission/reception point (TRP). In some configurations, various functions of each access network entity or base station **105** may be distributed across various network devices (e.g., radio heads and access network controllers) or consolidated into a single network device (e.g., a base station **105**).
(27) Wireless communications system **100** may operate using one or more frequency bands, which may be for example in the range of 300 megahertz (MHz) to 300 gigahertz (GHz). Generally, the region from 300 MHz to 3 GHz is known as the ultra-high frequency (UHF) region or decimeter band, since the wavelengths range from approximately one decimeter to one meter in length. UHF waves may be blocked or redirected by buildings and environmental features. However, the waves may penetrate structures sufficiently for a macro cell to provide service to UEs **115** located indoors. Transmission of UHF waves may be associated with smaller antennas and shorter range (e.g., less than 100 km) compared to transmission using the smaller frequencies and longer waves of the high frequency (HF) or very high frequency (VHF) portion of the spectrum below 300 MHz.
(28) Wireless communications system **100** may also operate in a super high frequency (SHF) region using frequency bands from 3 GHz to 30 GHz, also known as the centimeter band. The SHF region includes bands such as the 5 GHz industrial, scientific, and medical (ISM) bands, which

may be used opportunistically by devices that may be capable of tolerating interference from other users.
(29) Wireless communications system **100** may also operate in an extremely high frequency (EHF) region of the spectrum (e.g., from 30 GHz to 300 GHz), also known as the millimeter band. In some examples, wireless communications system **100** may support millimeter wave (mmW) communications between UEs **115** and base stations **105**, and EHF antennas of the respective devices may be even smaller and more closely spaced than UHF antennas. In some cases, this may facilitate use of antenna arrays within a UE **115**. However, the propagation of EHF transmissions may be subject to even greater atmospheric attenuation and shorter range than SHF or UHF transmissions. Techniques disclosed herein may be employed across transmissions that use one or more different frequency regions, and designated use of bands across these frequency regions may differ by country or regulating body.
(30) In some cases, wireless communications system **100** may utilize both licensed and unlicensed radio frequency spectrum bands. For example, wireless communications system **100** may employ License Assisted Access (LAA), LTE-Unlicensed (LTE-U) radio access technology, or NR technology in an unlicensed band such as the 5 GHz ISM band. When operating in unlicensed radio frequency spectrum bands, wireless devices such as base stations **105** and UEs **115** may employ listen-before-talk (LBT) procedures to ensure a frequency channel is clear before transmitting data. In some cases, operations in unlicensed bands may be based on a carrier aggregation configuration in conjunction with component carriers operating in a licensed band (e.g., LAA). Operations in unlicensed spectrum may include downlink transmissions, uplink transmissions, peer-to-peer transmissions, or a combination of these. Duplexing in unlicensed spectrum may be based on frequency division duplexing (FDD), time division duplexing (TDD), or a combination of both.
(31) In some examples, base station **105** or UE **115** may be equipped with multiple antennas, which may be used to employ techniques such as transmit diversity, receive diversity, multiple-input multiple-output (MIMO) communications, or beamforming. For example, wireless communications system **100** may use a transmission scheme between a transmitting device (e.g., a base station **105**) and a receiving device (e.g., a UE **115**), where the transmitting device is equipped with multiple antennas and the receiving device is equipped with one or more antennas. MIMO communications may employ multipath signal propagation to increase the spectral efficiency by transmitting or receiving multiple signals via different spatial layers, which may be referred to as spatial multiplexing. The multiple signals may, for example, be transmitted by the transmitting device via different antennas or different combinations of antennas. Likewise, the multiple signals may be received by the receiving device via different antennas or different combinations of antennas. Each of the multiple signals may be referred to as a separate spatial stream, and may carry bits associated with the same data stream (e.g., the same codeword) or different data streams. Different spatial layers may be associated with different antenna ports used for channel measurement and reporting. MIMO techniques include single-user MIMO (SU-MIMO) where multiple spatial layers are transmitted to the same receiving device, and multiple-user MIMO (MU-MIMO) where multiple spatial layers are transmitted to multiple devices.
(32) Beamforming, which may also be referred to as spatial filtering, directional transmission, or directional reception, is a signal processing technique that may be used at a transmitting device or a receiving device (e.g., a base station **105** or a UE **115**) to shape or steer an antenna beam (e.g., a transmit beam or receive beam) along a spatial path between the transmitting device and the receiving device. Beamforming may be achieved by combining the signals communicated via antenna elements of an antenna array such that signals propagating at particular orientations with respect to an antenna array experience constructive interference while others experience destructive interference. The adjustment of signals communicated via the antenna elements may include a transmitting device or a receiving device applying some amplitude and phase offsets to signals carried via each of the antenna elements associated with the device. The adjustments associated

with each of the antenna elements may be defined by a beamforming weight set associated with a particular orientation (e.g., with respect to the antenna array of the transmitting device or receiving device, or with respect to some other orientation).

(33) In one example, a base station **105** may use multiple antennas or antenna arrays to conduct beamforming operations for directional communications with a UE **115**. For instance, some signals (e.g., synchronization signals, reference signals, beam selection signals, or other control signals) may be transmitted by a base station **105** multiple times in different directions, which may include a signal being transmitted according to different beamforming weight sets associated with different directions of transmission. Transmissions in different beam directions may be used to identify (e.g., by the base station **105** or a receiving device, such as a UE **115**) a beam direction for subsequent transmission, or reception, or both by the base station **105**.

(34) Some signals, such as data signals associated with a particular receiving device, may be transmitted by a base station **105** in a single beam direction (e.g., a direction associated with the receiving device, such as a UE **115**). In some examples, the beam direction associated with transmissions along a single beam direction may be determined based on a signal that was transmitted in different beam directions. For example, a UE **115** may receive one or more of the signals transmitted by the base station **105** in different directions, and the UE **115** may report to the base station **105** an indication of the signal it received with a highest signal quality, or an otherwise acceptable signal quality. Although these techniques are described with reference to signals transmitted in one or more directions by a base station **105**, a UE **115** may employ similar techniques for transmitting signals multiple times in different directions (e.g., for identifying a beam direction for subsequent transmission or reception by the UE **115**), or transmitting a signal in a single direction (e.g., for transmitting data to a receiving device).

(35) A receiving device (e.g., a UE **115**, which may be an example of a mmW receiving device) may try multiple receive beams when receiving various signals from the base station **105**, such as synchronization signals, reference signals, beam selection signals, or other control signals. For example, a receiving device may try multiple receive directions by receiving via different antenna subarrays, by processing received signals according to different antenna subarrays, by receiving according to different receive beamforming weight sets applied to signals received at a plurality of antenna elements of an antenna array, or by processing received signals according to different receive beamforming weight sets applied to signals received at a plurality of antenna elements of an antenna array, any of which may be referred to as "listening" according to different receive beams or receive directions. In some examples, a receiving device may use a single receive beam to receive along a single beam direction (e.g., when receiving a data signal). The single receive beam may be aligned in a beam direction determined based on listening according to different receive beam directions (e.g., a beam direction determined to have a highest signal strength, highest signal-to-noise ratio, or otherwise acceptable signal quality based on listening according to multiple beam directions).

(36) In some cases, the antennas of a base station **105** or UE **115** may be located within one or more antenna arrays, which may support MIMO operations, or transmit or receive beamforming. For example, one or more base station antennas or antenna arrays may be co-located at an antenna assembly, such as an antenna tower. In some cases, antennas or antenna arrays associated with a base station **105** may be located in diverse geographic locations. A base station **105** may have an antenna array with a number of rows and columns of antenna ports that the base station **105** may use to support beamforming of communications with a UE **115**. Likewise, a UE **115** may have one or more antenna arrays that may support various MIMO or beamforming operations.

(37) In some cases, wireless communications system **100** may be a packet-based network that operate according to a layered protocol stack. In the user plane, communications at the bearer or Packet Data Convergence Protocol (PDCP) layer may be IP-based. A Radio Link Control (RLC) layer may perform packet segmentation and reassembly to communicate over logical channels. A

Medium Access Control (MAC) layer may perform priority handling and multiplexing of logical channels into transport channels. The MAC layer may also use hybrid automatic repeat request (HARQ) to provide retransmission at the MAC layer to improve link efficiency. In the control plane, the Radio Resource Control (RRC) protocol layer may provide establishment, configuration, and maintenance of an RRC connection between a UE **115** and a base station **105** or core network **130** supporting radio bearers for user plane data. At the Physical layer, transport channels may be mapped to physical channels.

(38) In some cases, UEs **115** and base stations **105** may support retransmissions of data to increase the likelihood that data is received successfully. HARQ feedback is one technique of increasing the likelihood that data is received correctly over a communication link **125**. HARQ may include a combination of error detection (e.g., using a cyclic redundancy check (CRC)), forward error correction (FEC), and retransmission (e.g., automatic repeat request (ARQ)). HARQ may improve throughput at the MAC layer in poor radio conditions (e.g., signal-to-noise conditions). In some cases, a wireless device may support same-slot HARQ feedback, where the device may provide HARQ feedback in a specific slot for data received in a previous symbol in the slot. In other cases, the device may provide HARQ feedback in a subsequent slot, or according to some other time interval.

(39) Time intervals in LTE or NR may be expressed in multiples of a basic time unit, which may, for example, refer to a sampling period of T.sub.s=1/30,720,000 seconds. Time intervals of a communications resource may be organized according to radio frames each having a duration of 10 milliseconds (ms), where the frame period may be expressed as T.sub.f=307,200 Ts. The radio frames may be identified by a system frame number (SFN) ranging from 0 to 1023. Each frame may include 10 subframes numbered from 0 to 9, and each subframe may have a duration of 1 ms. A subframe may be further divided into 2 slots each having a duration of 0.5 ms, and each slot may contain 6 or 7 modulation symbol periods (e.g., depending on the length of the cyclic prefix prepended to each symbol period). Excluding the cyclic prefix, each symbol period may contain **2048** sampling periods. In some cases, a subframe may be the smallest scheduling unit of the wireless communications system **100**, and may be referred to as a transmission time interval (TTI). In other cases, a smallest scheduling unit of the wireless communications system **100** may be shorter than a subframe or may be dynamically selected (e.g., in bursts of shortened TTIs (sTTIs) or in selected component carriers using sTTIs).

(40) In some wireless communications systems, a slot may further be divided into multiple mini-slots containing one or more symbols. In some instances, a symbol of a mini-slot or a mini-slot may be the smallest unit of scheduling. Each symbol may vary in duration depending on the subcarrier spacing or frequency band of operation, for example. Further, some wireless communications systems may implement slot aggregation in which multiple slots or mini-slots are aggregated together and used for communication between a UE **115** and a base station **105**.

(41) The term “carrier” refers to a set of radio frequency spectrum resources having a defined physical layer structure for supporting communications over a communication link **125**. For example, a carrier of a communication link **125** may include a portion of a radio frequency spectrum band that is operated according to physical layer channels for a given radio access technology. Each physical layer channel may carry user data, control information, or other signaling. A carrier may be associated with a pre-defined frequency channel (e.g., an evolved universal mobile telecommunication system terrestrial radio access (E-UTRA) absolute radio frequency channel number (EARFCN)), and may be positioned according to a channel raster for discovery by UEs **115**. Carriers may be downlink or uplink (e.g., in an FDD mode), or be configured to carry downlink and uplink communications (e.g., in a TDD mode). In some examples, signal waveforms transmitted over a carrier may be made up of multiple sub-carriers (e.g., using multi-carrier modulation (MCM) techniques such as orthogonal frequency division multiplexing (OFDM) or discrete Fourier transform spread OFDM (DFT-S-OFDM)).

(42) The organizational structure of the carriers may be different for different radio access technologies (e.g., LTE, LTE-A, LTE-A Pro, NR). For example, communications over a carrier may be organized according to TTIs or slots, each of which may include user data as well as control information or signaling to support decoding the user data. A carrier may also include dedicated acquisition signaling (e.g., synchronization signals or system information, or the like) and control signaling that coordinates operation for the carrier. In some examples (e.g., in a carrier aggregation configuration), a carrier may also have acquisition signaling or control signaling that coordinates operations for other carriers.
(43) Physical channels may be multiplexed on a carrier according to various techniques. A physical control channel and a physical data channel may be multiplexed on a downlink carrier, for example, using time division multiplexing (TDM) techniques, frequency division multiplexing (FDM) techniques, or hybrid TDM-FDM techniques. In some examples, control information transmitted in a physical control channel may be distributed between different control regions in a cascaded manner (e.g., between a common control region or common search space and one or more UE-specific control regions or UE-specific search spaces).
(44) A carrier may be associated with a particular bandwidth of the radio frequency spectrum, and in some examples, the carrier bandwidth may be referred to as a "system bandwidth" of the carrier or the wireless communications system **100**. For example, the carrier bandwidth may be one of a number of predetermined bandwidths for carriers of a particular radio access technology (e.g., 1.4, 3, 5, 10, 15, 20, 40, or 80 MHz). In some examples, each served UE **115** may be configured for operating over portions or all of the carrier bandwidth. In other examples, some UEs **115** may be configured for operation using a narrowband protocol type that is associated with a predefined portion or range (e.g., set of subcarriers or RBs) within a carrier (e.g., "in-band" deployment of a narrowband protocol type).
(45) In a system employing MCM techniques, a resource element may include or consist of one symbol period (e.g., a duration of one modulation symbol) and one subcarrier, where the symbol period and subcarrier spacing are inversely related. The number of bits carried by each resource element may depend on the modulation scheme (e.g., the order of the modulation scheme). Thus, the more resource elements that a UE **115** receives and the higher the order of the modulation scheme, the higher the data rate may be for the UE **115**. In MIMO systems, a wireless communications resource may refer to a combination of a radio frequency spectrum resource, a time resource, and a spatial resource (e.g., spatial layers), and the use of multiple spatial layers may further increase the data rate for communications with a UE **115**.
(46) Devices of the wireless communications system **100** (e.g., base stations **105** or UEs **115**) may have a hardware configuration that supports communications over a particular carrier bandwidth, or may be configurable to support communications over one of a set of carrier bandwidths. In some examples, the wireless communications system **100** may include base stations **105** or UEs **115**, or both that support simultaneous communications via carriers associated with more than one different carrier bandwidth. Wireless communications system **100** may support communication with a UE **115** on multiple cells or carriers, a feature which may be referred to as carrier aggregation or multi-carrier operation. A UE **115** may be configured with multiple downlink component carriers and one or more uplink component carriers according to a carrier aggregation configuration. Carrier aggregation may be used with both FDD and TDD component carriers.
(47) In some cases, wireless communications system **100** may utilize enhanced component carriers (eCCs). An eCC may be characterized by one or more features including wider carrier or frequency channel bandwidth, shorter symbol duration, shorter TTI duration, or modified control channel configuration. In some cases, an eCC may be associated with a carrier aggregation configuration or a dual connectivity configuration (e.g., when multiple serving cells have a suboptimal or non-ideal backhaul link). An eCC may also be configured for use in unlicensed spectrum or shared spectrum (e.g., where more than one operator is allowed to use the spectrum). An eCC characterized by wide

carrier bandwidth may include one or more segments that may be utilized by UEs **115** that are not capable of monitoring the whole carrier bandwidth or are otherwise configured to use a limited carrier bandwidth (e.g., to conserve power).

(48) In some cases, an eCC may utilize a different symbol duration than other component carriers, which may include use of a reduced symbol duration as compared with symbol durations of the other component carriers. A shorter symbol duration may be associated with increased spacing between adjacent subcarriers. A device, such as a UE **115** or base station **105**, utilizing eCCs may transmit wideband signals (e.g., according to frequency channel or carrier bandwidths of 20, 40, 60, 80 MHz, etc.) at reduced symbol durations (e.g., 16.67 microseconds). A TTI in eCC may include or consist of one or multiple symbol periods. In some cases, the TTI duration (that is, the number of symbol periods in a TTI) may be variable. Wireless communications system **100** may be an NR system that may utilize any combination of licensed, shared, and unlicensed spectrum bands, among others. The flexibility of eCC symbol duration and subcarrier spacing may allow for the use of eCC across multiple spectrums. In some examples, NR shared spectrum may increase spectrum utilization and spectral efficiency, specifically through dynamic vertical (e.g., across the frequency domain) and horizontal (e.g., across the time domain) sharing of resources.

(49) A base station **105** may perform a connection procedure, such as a cell acquisition procedure, a random access procedure, or the like with a UE **115**. For example, a base station **105** and a UE **115** may perform a random access procedure to establish a connection. In some other examples, a base station **105** and a UE **115** may perform a random access procedure to re-establish a connection after connection failure (such as radio-link failure) with the base station **105**, or to establish a connection for handover to another base station, or the like. In some examples, the random access procedure may be a four-step random access procedure. As part of the four-step random access procedure, the UE **115** may transmit a random access message (message 1 (msg1)) carrying a random access preamble. The random access preamble may enable the base station **105** to distinguish between multiple UEs **115** attempting to access the wireless communications system **100** simultaneously.

(50) The base station **105** may respond with a random access response (message 2 (msg2)) that provides an uplink resource grant, a timing advance, and a temporary cell-radio network temporary identifier (C-RNTI). The UE **115** may transmit a subsequent random access message (message 3 (msg3)) that includes an RRC connection request along with a temporary mobile subscriber identity (TMSI) (e.g., if the UE **115** has previously been connected to the same wireless network) or a random identifier. The RRC connection request may also indicate the reason the UE **115** is connecting to the network, for example, such as one or more of an emergency, signaling, or data exchange. The base station **105** may respond to the connection request with a contention resolution message (message 4 (msg4)) addressed to the UE **115**, which may provide a new C-RNTI. If the UE **115** receives a contention resolution message with the correct identification, it may proceed with an RRC connection setup. If the UE **115** does not, however, receive a contention resolution message (for example, if there is a conflict with another UE **115**) it may repeat the random access process by transmitting a new random access preamble. As described, the exchange of messages between the UE **115** and the base station **105** for random access may be referred to as a four-step random access procedure.

(51) In other examples, a two-step random access procedure may be performed for random access. A UE **115** that operates in licensed or unlicensed spectrum within the wireless communications system **100** may participate in a two-step random access procedure to reduce delay in establishing communication with a base station **105** (such as compared to a four-step random access procedure). In some examples, the two-step random access procedure may operate regardless of whether a UE **115** has a valid timing advance parameter. For example, the UE **115** may use a valid timing advance parameter to coordinate the timing of its transmissions to a base station **105** (to account for propagation delay) and may receive the valid timing advance parameter as part of the two-step random access procedure. The two-step random access procedure may be applicable to any cell

size regardless of whether the random access procedure is contention-based or contention-free, and may combine multiple random access messages from a four-step random access procedure.
(52) The UE **115** may transmit a first random access message (for example, message A (msgA)) to the base station **105**. The first random access message may include, for example, the contents of a msgl and a msg3 from a four-step random access procedure. Additionally, msgA may include or consist of a random access preamble and a PUSCH carrying a random access payload with the contents of the message (for example, the content of a msg3 in a four-step random access procedure). The UE **115** may transmit the random access preamble on a physical random access channel (PRACH). In some examples, the PUSCH may also carry a reference signal, such as a PUSCH demodulation reference signal (DMRS). In some other examples, the base station **105** may transmit a downlink control channel (such as a physical downlink control channel (PDCCH) or physical downlink shared channel (PDSCH)) carrying a corresponding second random access message (for example, message B (msgB)) to the UE **115**. The second message may include, for example, the contents of a msg2 and a msg4 from a four-step random access procedure.
(53) A random access payload of a PUSCH may, in some examples, have a variable payload size, for example, based on contents of a random access message (for example, contents of a msg3 in a four-step random access procedure). The random access payload may have a baseline payload size (which may also be referred to as a default payload size), for example, such as 56 bits or 72 bits. In some examples, however, a payload size of the random access payload may exceed resources allocated for the PUSCH. As demand for communication efficiency increases in the wireless communications system **100**, it may be desirable for UEs **115** to target efficient resource usage for random access procedures, particularly for variable payload sizes of random access payloads carried on PUSCHs. As described herein, base stations **105** and UEs **115** may support aggregating PUSCH resource units relating to a random access message (e.g., msgA) in a random access procedure, for example, to address challenges with variable size random access payloads for PUSCH transmissions. As a result, UEs **115** may be capable of supporting larger payload sizes of random access payloads, as well as reduce a modulation and coding scheme of random access messages. For example, each PUSCH resource unit may be capable of supporting a predefined payload, for example, such as 56 bits, 72 bits, 1000 bits, or the like with a modulation and coding scheme as further described in more detail herein.
(54) Base stations **105** may also include a base station communications manager **102**, which may manage communications in a terrestrial communications system. For a base station **105**, base station communications manager **102** may assign a set of PUSCH resource units associated with one or more POs for transmission of a random access payload of a random access message associated with a random access procedure. In some examples, base station communications manager **102** may determine an aggregation configuration for multiple PUSCH resource units of the set of PUSCH resource units based on assigning the set of PUSCH resource units. The base station communications manager **102** may transmit, for example, to the UE communications manager **101** of a UE **115**, an indication of the aggregation configuration for the multiple PUSCH resource units.
(55) UEs **115** may include a UE communications manager **101**, which may manage communications in a terrestrial communications system. For a UE **115**, UE communications manager **101** may identify a random access message (e.g., msgA) of a random access procedure (e.g., a two-step random access procedure). The random access message may include a random access preamble and a random access payload. The UE communications manager **101** may aggregate multiple PUSCH resource units of a set of PUSCH resource units associated with one or more POs based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload.
(56) In some examples, transmission of a random access preamble on a PRACH and transmission of a PUSCH carrying a random access payload may occur over one or more occasions. For

example, transmission of a random access preamble on a PRACH may occur over a random access channel occasion (RO), which may include time and frequency resources. In other examples, transmission of a PUSCH carrying a random access payload may occur over a PO, which may include time and frequency resources. In some examples, the time and frequency resources may be consecutive or non-consecutive. There may also be a relationship between one or more occasions, for example, ROs and POs may have a one-to-one mapping. That is, a single RO may map to a single respective PO. In some examples, ROs and POs may have one-to-many mappings, for example, a single RO may map to several POs. Alternatively, multiple ROs may map to a single PO using a many-to-one mapping. The UE **115** may multiplex bits of a random access payload over time and frequency resources of the aggregated multiple PUSCH resource units of one or more POs. The UE communications manager **101** may transmit, to base station communications manager **102**, the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs. The wireless communications system **100** may therefore include features for improved resource usage for random access messages, improved reliability for random access messaging and, in some examples, may promote low latency for random access procedures, among other benefits.
(57) FIG. **2**A illustrates an example of a wireless communications system **200**-*a* that supports aggregated uplink shared channel transmission for two-step random access channel procedures in accordance with one or more aspects of the present disclosure. In some examples, the wireless communications system **200**-*a* may implement aspects of the wireless communications system **100**. For example, the wireless communications system **200**-*a* may include a base station **105**-*a* and a UE **115**-*a*, which may be examples of the corresponding devices described with reference to FIG. **1**.
(58) The base station **105**-*a* may perform a connection procedure (e.g., an RRC procedure, a cell acquisition procedure, a random access procedure) with the UE **115**-*a*. For example, the base station **105**-*a* and the UE **115**-*a* may perform a random access procedure to establish a connection for wired or wireless communication. In other examples, the base station **105**-*a* and the UE **115**-*a* may perform a random access procedure to re-establish a connection after an issue, such as connection failure, with the base station **105**-*a*, or the like. The base station **105**-*a* and the UE **115**-*a* may also support multiple radio access technologies including 4G systems such as LTE systems, LTE-A systems, or LTE-A Pro systems, and 5G systems which may be referred to as NR systems.
(59) The connection procedure (e.g., random access procedure) between the base station **105**-*a* and the UE **115**-*a* may correspond to, for example, at least one of the above example radio access technologies. By way of example, in FIG. **2**A, a random access procedure may be related to 4G systems and may be referred to as a four-step random access procedure. As part of the four-step random access procedure, the base station **105**-*a* and the UE **115**-*a* may transmit one or more messages (e.g., handshake messages), such as a random access message **205** (which may also be referred to as msg1), a random access message **210** (which may also be referred to as msg2), a random access message **215** (which may also be referred to as msg3), and a random access message **220** (which may also be referred to as msg4).
(60) The random access procedure may include the UE **115**-*a* transmitting the random access message **205**, which may include a preamble (which may also be referred to as a random access channel preamble, a PRACH preamble, or a sequence) that may carry information, such as a UE identifier. In some examples, the UE **115**-*a* may transmit a preamble over an RO. The purpose of the preamble transmission may be to provide an indication to the base station **105**-*a* presence of a random access attempt, and to allow the base station **105**-*a* to determine a delay (e.g., a timing delay) between the base station **105**-a and the UE **115**-*a*. The UE **115**-*a* may transmit the random access message **205** to the base station **105**-*a* on a PRACH, for example.
(61) The preamble of the random access message **205** may, in some examples, be defined by a preamble sequence and a cyclic prefix. A preamble sequence may be defined based on a Zadoff-

Chu sequence. The UE **115-***a* may additionally or alternatively, use a guard period to handle timing uncertainty of the random access message **205** transmission. For example, before beginning the random access procedure, the UE **115-***a* may obtain downlink synchronization with the base station **105-***a* based in part on a cell-search procedure.

(62) However, because the UE **115-***a* has not yet obtained uplink synchronization with the base station **105-***a*, there may be an uncertainty in uplink timing due to the location of the UE **115-***a* in the cell (e.g., geographic coverage area of base station **105-***a*) not being known.

(63) In some examples, the uncertainty in uplink timing may be based on a dimension (e.g., size, area) of the cell. Therefore, including a cyclic prefix to the random access message **205** may be beneficial, in some examples, for handling the uncertainty in uplink timing. Per cell, there may be a number of preamble sequences (e.g., 64 preamble sequences). The UE **115-***a* may select a preamble sequence from a set of sequences in a cell (e.g., geographic coverage area of base station **105-***a*) based on, for example, a randomness selection. In some examples, the UE **115-***a* may select a preamble sequence based on an amount of traffic that the UE **115-***a* has for transmission on an uplink shared channel (UL-SCH). From the preamble sequence that the UE **115-***a* selected, the base station **105-***a* may determine the amount of uplink resources to be granted to the UE **115-***a*.

(64) Some examples of a random access procedure may be contention-based or contention-free. When performing a contention-based random access (CBRA) procedure, the UE **115-***a* may select a preamble sequence from a set of sequences. That is, as long as other UEs (not shown) are not performing a random access attempt using the same sequence at a same temporal instance, no collisions will occur, and the random access attempt may be detected by the base station **105-***a*. If the UE **115-***a* is performing a contention-free random access (CFRA) attempt, for example, for a handover to a new cell, the preamble sequence to use may be explicitly signaled (e.g., in control information) by the base station **105-***a*. To avoid collisions or interference, the base station **105-***a* may select a contention-free preamble sequence from sequences not associated with the CBRA attempt.

(65) Upon receiving the random access message **205**, the base station **105-***a* may respond appropriately with a random access message **210**. For example, the base station **105-***a* may transmit the random access message **210** to the UE **115-***a* on a downlink shared channel (DL-SCH) or a PDCCH. In some examples, the random access message **210** may have a same or a different configuration (format) compared to the random access message **205**. The random access message **210** may carry information for the UE **115-***a*, where the information may be determined by the base station **105-***a* based on information carried in the random access message **205**. For example, the information in the random access message **210** may include an index of a detected preamble sequence and for which the response is valid, a timing advance parameter determined based on the detected preamble sequence, a scheduling grant indicating time and frequency resources for the UE **115-***a* to use for transmission of a next random access message transmission by the UE **115-***a*, or a network identifier for further communication with the UE **115-***a*, or the like.

(66) The UE **115-***a* (and additional UEs (not shown)) may monitor the PDCCH to detect and receive a random access message (e.g., the random access message **210**), from the base station **105-***a*. In some examples, the UE **115-***a* may monitor the PDCCH for a random access message transmission from the base station **105-***a* during a random access response (RAR) window, which may be fixed or variable in size. For example, if the UE **115-***a* does not detect and receive a random access message transmission from the base station **105-***a*, the random access attempt may be declared as a failure and the random access procedure in FIG. **2**A may repeat. However, in the subsequent attempt, the RAR window may be adjusted (e.g., increased or decreased in duration). In some examples, the base station **105-***a* may configure an expected time window for the RAR window and an expiration timer for random access message retransmissions.

(67) Once the UE **115-***a* successfully receives the random access message **210**, the UE **115-***a* may obtain uplink synchronization with the base station **105-***a*. In some examples, before data

transmission from the UE **115**-*a*, a unique identifier within the cell may be assigned to the UE **115**-*a*. In some examples, depending on a state (e.g., a connected state, an ideal state) of the UE **115**-*a* there may be a desire for an additional message (e.g., a connection request message) exchange for setting up the connection between the base station **105**-*a* and the UE **115**-*a*. The UE **115**-*a* may transmit any messages, for example, the random access message **215** to the base station **105**-*a* using the UL-SCH resources (or PUSCH resources) assigned in the random access message **210**. The random access message **210** may include a UE identifier for contention resolution. If the UE **115**-*a* is in a connected state, for example, the UE identifier may be a C-RNTI. Otherwise, the UE identifier may be specific to the UE **115**-*a*.

(68) The base station **105**-*a* may receive the random access message **215** and may respond properly, for example, by transmitting the random access message **220**, which may be a contention resolution message. When multiple UEs (including the UE **115**-*a*) are performing simultaneously random access attempts using a same preamble sequence, these UEs may result in listening for a same response message (e.g., the random access message **220**). Each UE (including the UE **115**-*a*) may receive the random access message **220** and compare an identifier (e.g., network identifier) in the random access message **220** to the identifier specified in the random access message **215**. When the identifiers match, the corresponding UE (e.g., the UE **115**-*a*) may declare the random access procedure successful. UEs that do not identify a match between the identifiers are considered to have failed the random access procedure and may repeat the random access procedure with the base station **105**-*a*. As a result of the connection procedure, the base station **105**-*a* and the UE **115**-*a* may establish a connection for wired or wireless communication. Although, the connection procedure (e.g., random access procedure) in FIG. **2**A may be effective for facilitating random access for the UE **115**-*a*, there may be unnecessary latencies associated with this procedure. The UE **115**-*a* may therefore support a two-step random access procedure to reduce latencies associated with processes related to initial channel access.

(69) FIG. **2**B illustrates an example of a wireless communications system **200**-*b* that supports aggregated uplink shared channel transmission for two-step random access channel procedures in accordance with one or more aspects of the present disclosure. In some examples, the wireless communications system **200**-*b* may implement aspects of the wireless communications system **100**. For example, the wireless communications system **200**-*b* may include a base station **105**-*a* and a UE **115**-*a*, which may be examples of the corresponding devices described with reference to FIG. **1**. The wireless communications system **200**-*b* may support variable payload sizes and, in some examples, may promote low latency of random access procedures, among other benefits.

(70) The base station **105**-*a* and the UE **115**-*a* may, as explained with reference to FIG. **2**A, support multiple radio access technologies including 4G systems such as LTE systems, LTE-A systems, or LTE-A Pro systems, and 5G systems which may be referred to as NR systems. The connection procedure (e.g., random access procedure) between the base station **105**-*a* and the UE **115**-*a* may correspond to, for example, at least one of these example radio access technologies. In the example of FIG. **2**B, the random access procedure may be related to 5G systems and may be referred to as a two-step random access procedure. As part of a two-step random access procedure, to decrease latencies related to contention-based aspects of the two-step random access procedure, the base station **105**-*a* and the UE **115**-*a* may exchange fewer messages (e.g., handshake messages) compared to a four-step random access procedure, as explained in FIG. **2**A.

(71) For example, the UE **115**-*a* may transmit a single message, such as a random access message **225** (which may also be referred to as msgA), and the base station **105**-*a* may transmit a single message, such as a random access response message **230** (which may also be referred to as msgB) in response to the random access message **225**. The random access message **225** (e.g., msgA) may combine parts of msgs1,3 of a four-step random access procedure, while the random access response message **230** (e.g., msgB) may combine aspects of msgs2,4 of the four-step random access procedure.

(72) In some examples, transmission of a random access preamble on a PRACH and transmission of a PUSCH carrying a random access payload may occur over one or more occasions. For example, transmission of a random access preamble on a PRACH may occur over an RO, which may include time and frequency resources. In other examples, transmission of a PUSCH carrying a random access payload may occur over a PO, which may include time and frequency resources. In some examples, the time and frequency resources may be consecutive or non-consecutive. There also may be a relationship between one or more occasions, for example, ROs and POs may have a one-to-one mapping. That is, a single RO may map to a single respective PO. In some examples, ROs and POs may have one-to-many mappings, for example, a single RO may map to several POs. Alternatively, multiple ROs may map to a single PO using a many-to-one mapping.
(73) In some examples, a random access payload of the random access message **225** (e.g., msgA) may have a variable payload size, for example, based on contents of a random access message (for example, contents of a msg3 in a four-step random access procedure). In some examples, however, a payload size of the random access payload may exceed resources allocated for the PUSCH. It may be therefore desirable for UE **115**-*a* to target improved resource usage for random access procedures, particularly for variable payload sizes of random access payloads carried on PUSCHs of the random access message **225** (e.g., msgA). With reference to FIGS. **2**A and **2**B, the base station **105**-*a* and the UE **115**-*a* may support aggregating PUSCH resource units relating to a random access message (e.g., msgA) in a random access procedure, to address challenges with variable size random access payloads.
(74) As a result, the base station **105**-*a* and the UE **115**-*a* may be capable of supporting larger payload sizes of random access payloads, as well as reducing a modulation and coding scheme of random access messages. For example, PUSCH resource units may be capable of supporting variable payload size, for example, such as 8 bits, 16 bits, 24 bits, 56 bits, 72 bits, 1000 bits, or the like with a modulation and coding scheme.
(75) The base station **105**-*a* may assign a set of PUSCH resource units associated with one or more POs for transmission of a random access payload of the random access message **225** (e.g., msgA), and determine an aggregation configuration for multiple PUSCH resource units of the set of PUSCH resource units. A PUSCH resource unit may occupy time and frequency resources. Time and frequency resources, as described herein, may include symbol periods, slots, subcarriers, carriers, and the like. Each PUSCH resource unit may therefore span a number of symbols, slots, and physical resource blocks, and may carry a predefined payload (e.g., 56 bits or 72 bits).
(76) In some examples, a PUSCH resource unit of the set of PUSCH resource units may span a PO, which may also include an allocation of time and frequency resources for a PUSCH transmission. In some other examples, multiple aggregated PUSCH resource units may span over multiple POs. PUSCH resource units of the multiple aggregated PUSCH resource units may be, in some examples, contiguous in one or more of a time domain or a frequency domain. Additionally or alternatively, PUSCH resource units of the multiple aggregated PUSCH resource units may be, in some other examples, non-contiguous in one or more of a time domain or a frequency domain.
(77) In some examples, the UE **115**-*a* may transmit a modulation and coding scheme capability, which the base station **105**-*a* may use to determine an aggregation configuration for multiple PUSCH resource units. In some examples, the base station **105**-*a* may identify that the UE **115**-*a* supports a single modulation and coding scheme based on the provided modulation and coding scheme capability. Multiple aggregated PUSCH resource units may, in some examples, use a same modulation and coding scheme. In some examples, if the modulation and coding scheme capability supports a single modulation and coding scheme (e.g., quadrature phase shift keying (QPSK) plus half coding rate), a payload size of one or more PUSCH resource units may be fixed. Alternatively, if the modulation and coding scheme capability supports multiple modulation and coding schemes, a payload size of one or more PUSCH resource units may be fixed or variable.
(78) For a variable payload size of one or more PUSCH resource units, the UE **115**-*a* may transmit

an indication of occupied resources (e.g., resource elements, resource element groups) in a preamble or in a dedicated control part of the random access message **225** (e.g., msgA) to the base station **105**-*a*. In some examples, the UE **115**-*a* may apply a zero padding on unused bits of a random access payload when a payload size of the random access payload is smaller than a payload size of a PUSCH resource unit. For example, the UE **115**-*a* may compare a payload size of a random access payload of the random access message **225** to an aggregated payload size of an aggregated multiple PUSCH resource unit, and determine a number of unused bits of the aggregated payload size of the aggregated multiple PUSCH resource units based on the comparison. As a result, the UE **115**-*a* may pad each unused bit of the number of unused bits of the aggregated payload size of the aggregated multiple PUSCH resource units with a null bit.

(79) The base station **105**-*a* may transmit the aggregation configuration for multiple PUSCH resource units to the UE **115**-*a* via higher layer signaling, for example, such as RRC signaling. The UE **115**-*a* may aggregate multiple PUSCH resource units of a set of PUSCH resource units associated with one or more POs based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload. The UE **115**-*a* may transmit the random access payload of the random access message **225** on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over one or more POs.

(80) In some examples, base station **105**-*a* may, as part of the random access procedure, transmit an RO configuration or a PO configuration to the UE **115**-*a*. The RO configuration may include one or more of: a periodicity of an RO, a time and frequency resource allocation of the RO, a preamble sequence configuration for a CBRA procedure or a CFRA procedure, an association rule between ROs and POs, or a beam association rule between a synchronization signal block (SSB) or a channel state information reference signal (CSI-RS) and the RO. The PO configuration may include one or more of: a periodicity of a PO, a time and frequency resource allocation of the PO or a PUSCH resource unit, a DMRS configuration, or a PUSCH waveform configuration, beam management information for a PUS CH.

(81) The UE **115**-*a* may use the association rule between ROs and POs provided in the RO configuration to support mapping between ROs and POs. In some examples, ROs and POs may have a one-to-one mapping. For example, a single RO may map to a single PO. In some examples, ROs and POs may have one-to-many mappings, for example, a single RO may map to several POs. Alternatively, multiple ROs may map to a single PO using a many-to-one mapping. Therefore, the UE **115**-*a* may support different mapping configurations (e.g., one-to-many mappings) between ROs and POs using the association rule. In some examples, the association rule may be preconfigured by a network operator. In some examples, base station **105**-*a* may transmit the association rule in the RO configuration via upper layer signaling (e.g., RRC signaling).

(82) The UE **115**-*a* may map one or more ROs to one or more POs to support the multiple aggregated PUSCH resource units. In some examples, one or more of the one or more ROs or the one or more POs may be noncontiguous in one or more of a time domain or a frequency domain. Alternatively, one or more of the one or more ROs or the one or more POs may be contiguous in one or more of the time domain or the frequency domain. In some examples, UE **115**-*a* may map the random access preamble of the random access message **225** to the one or more POs in one or more of the time domain or the frequency domain based on the RO configuration or the PO configuration.

(83) In some other examples, the UE **115**-*a* may assign, to the one or more POs, available random access preambles including the random access preamble of the random access message **225** associated with one or more ROs, and map the assigned random access preambles to the one or more POs. For example, the UE **115**-*a* may divide available preambles per RO to multiple sets, where each set or specific sets are associated with one or more POs or one or more PUSCH resource units (e.g., a first preamble set may map to a first PUSCH resource unit (or PO) and a

second preamble set may map to multiple PUSCH resource units (or multiple POs), and the like). In some examples, the UE **115-*a*** may aggregate multiple PUSCH resource units based one or more factors, for example, such as a latency or a transmit power.

(84) For example, the UE **115-*a*** may map POs to ROs first in a frequency domain and second in a time domain to reduce latency. Alternatively, the UE **115-*a*** may map POs to ROs first in a time domain and second in a frequency domain to increase transmit power. The UE **115-*a*** may map the random access payload of the random access message **225** to one or more of the one or more ROs or the one or more POs in one or more of the time domain or the frequency domain based on a hopping sequence (e.g., a timing hopping sequence, a frequency hopping sequence, or a timing and frequency hopping sequence).

(85) The UE **115-*a*** may, in some examples, map a DMRS or a set of DMRS to random access preambles of PUSCH resource units of the aggregated multiple PUSCH resource units based on a mapping rule between one or more DMRS and preambles. In some examples, each PUSCH resource unit of the aggregated multiple PUSCH resource units may be associated with a unique DMRS or a set of DMRS. Alternatively, the aggregated multiple PUSCH resource units may share a same DMRS or set of DMRS. The base station **105-*a*** may use the DMRS or the set of DMRS to successfully receive and decode the preamble and random access payload of the random access message **225**.

(86) In some examples, the UE **115-*a*** may transmit the random access payload of the random access message **225** on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over one or more POs based on a power configuration. For example, the base station **105-*a*** may transmit an indication (e.g., via RRC signaling) of a power configuration that may include a transmit power for the UE **115-*a*** to use when transmitting the random access payload of the random access message **225** on the PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over one or more POs. In an example, the power configuration may indicate for UE **115-*a*** to use a same transmit power for the aggregated multiple PUSCH resource units.

(87) When supporting both two-step and four-step random access procedures, in some examples, the base station **105-*a*** and the UE **115-*a*** may commence with one random access procedure (e.g., a two-step random access procedure) and fallback to another random access procedure (e.g., a four-step random access procedure), if needed. The base station **105-*a*** may select or assign a priority to a random access procedure based in part on a parameter (e.g., traffic type, network load).

(88) For example, the base station **105-*a*** may configure to use a four-step random access procedure over a two-step random access procedure for some scenarios (e.g., traffic load) to reduce an overhead because the base station **105-*a*** may have to provision for msgA resources for two-step random access procedures that may involve a larger overhead compared to msgl transmissions with four-step random access procedures.

(89) Additionally, or alternatively, the base station **105-*a*** may select a random access procedure based on a UE **115-*a*** capability to support the random access procedure. For example, if UE **115-*a*** supports both two-step and four-step random access procedures, the base station **105-*a*** may select the two-step or four-step random access procedure to commence the initial access procedure. Otherwise, the base station **105-*a*** may select the random access procedure supported by the UE **115-*a***.

(90) Returning to the example of random access messaging, the UE **115-*a*** may transmit the random access message **225** to the base station **105-*a***. The random access message **225** may include a preamble and a PUSCH carrying a random access payload using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs, where information in the random access message **225** (e.g., msgA) includes the equivalent contents or aspects of random access message **215** (e.g., msg**3** of the four-step random access procedure). An advantage of the two-step random access procedure compared to the four-step random access procedure is that the

UE **115**-*a* may be capable of transmitting data (e.g., payload in PUSCH) to the base station **105**-*a* without having to be in a connected state for one data transmission.
(91) The base station **105**-*a* may monitor a PUSCH for a random access preamble or a random access payload of the random access message **225**. The random access payload may carry a connection request. In some examples, the base station **105**-*a* may determine an absence of the random access preamble or the random access payload of the random access message **225** based on the monitoring. Absence of the random access preamble or the random access payload of the random access message **225** may result in a random access procedure failure.
(92) In some examples, if the preamble of the random access message **225** is detected, the base station **105**-*a* may transmit a random access response message including an indication to perform a random access fallback procedure (e.g., four-step random access procedure). The UE **115**-*a* may perform the random access fallback procedure based on receiving the random access response message. In some examples, the base station **105**-*b* may request the UE **115**-*a* to retransmit the random access message **225**, instead of performing the random access fallback procedure. Alternatively, after successfully receiving the random access message **225**, the base station **105**-*a* may construct and transmit the random access response message **230** to the UE **115**-*a*. For example, the base station **105**-*a* may transmit the random access response message **230** to the UE **115**-*a* on a DL-SCH, PDSCH, PDCCH.
(93) The wireless communications systems **200** may support improvements and extending resource usage (e.g., time and frequency resources) for random access messages, among other advantages. As such, supported techniques may include features for efficient random access procedures. The described techniques may also support improved reliability for random access messaging and, in some examples, may promote low latency for random access procedures, among other benefits.
(94) FIG. **3** illustrates an example of a process flow **300** that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure. The process flow **300** may implement aspects of the wireless communications systems **100** and **200**, as described with reference to FIGS. **1**, **2**A, and **2**B. For example, the process flow **300** may be based on a configuration by a base station **105**, and implemented by a UE **115** for aggregating multiple PUSCH resources units to support larger payload sizes of random access messages or reduce a modulation and coding scheme of the random access messages, as described herein.
(95) The process flow **300** may include a base station **105**-*b* and a UE **115**-*b*, which may be examples of a base station **105** and a UE **115** as described with reference to FIG. **1**. In the following description of the process flow **300**, the operations between the base station **105**-*b* and the UE **115**-*b* may be transmitted in a different order than the example order shown, or the operations performed by the base station **105**-*b* and the UE **115**-*b* may be performed in different orders or at different times. Some operations may also be omitted from the process flow **300**, and other operations may be added to the process flow **300**. In the example of FIG. **3**, the base station **105**-*b* and the UE **115**-*b* may be in communication with each other via a terrestrial network. The process flow **300** may support higher data rates, improved random access messaging support for the UE **115**-*b*, among other benefits.
(96) In some examples, the process flow **300** may be part of a random access procedure to establish a connection between the base station **105**-*b* and the UE **115**-*b*. The base station **105**-*b* and the UE **115**-*b* may support multiple radio access technologies including 4G systems such as LTE systems, LTE-A systems, or LTE-A Pro systems, and 5G systems which may be referred to as NR systems. The random access procedure may correspond to, for example, at least one of the above example radio access technologies. In FIG. **3**, by way of example, the random access procedure may be a two-step or four-step random access procedure related to 4G or 5G systems.
(97) At **305**, the base station **105**-*b* may assign a set of PUSCH resource units associated with one or more POs for transmission of a random access message, which may include a random access

preamble and a random access payload. At **310**, the base station **105**-*b* may determine an aggregation configuration for multiple PUSCH resource units based on the assignment. At **315**, the base station **105**-*b* may transmit the aggregation configuration to the UE **115**-*b*, for example, via RRC signaling.

(98) At **320**, the UE **115**-*b* may aggregate multiple PUSCH resource units. For example, the UE **115**-*b* may aggregate, using the aggregation configuration, multiple PUSCH resource units of a set of PUSCH resource units associated with one or more POs based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload. At **325**, the UE **115**-*b* may transmit the random access payload of the random access message (e.g., msgA) on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs.

(99) At **330**, the base station **105**-*b* may transmit a random access response message (which may also be referred to as msgB) to the UE **115**-*b*. The msgB may be a random access response to the received msgA, including a random access preamble and the random access payload from the UE **115**-*b*. In some examples, the base station **105**-*b* may transmit the msgB to the UE **115**-*b* based on a UE contention resolution identifier, a random access radio network temporary identifier (RA-RNTI), or the like. For example, as part of the random access procedure, the base station **105**-*b* may transmit the msgB on a PDCCH or PDSCH according to the RA-RNTI.

(100) At **335**, the process flow **300** may proceed with the base station **105**-*b* and the UE **115**-*b* establishing the connection. At **340**, the process flow **300** may proceed with the base station **105**-*b* and the UE **115**-*b* communicating uplink and downlink communications, for example, such as control information, data, and the like.

(101) The operations performed by the base station **105**-*b* and the UE **115**-*b* as part of, but not limited to, process flow **300** may provide improvements to UE **115**-*b* resource usage and allocation for random access messages. Further, the operations performed by the base station **105**-*b* and the UE **115**-*b* as part of, but not limited to, process flow **300** may provide benefits and enhancements to the operation of the UE **115**-*b*. For example, the described aggregation of multiple PUSCH resources units in the process flow **300** may support improved data rates and enhanced random access messaging reliability, among other advantages.

(102) FIG. **4** shows a block diagram **400** of a device **405** that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure. The device **405** may be an example of aspects of a UE **115** as described herein. The device **405** may include a receiver **410**, a UE communications manager **415**, and a transmitter **420**. The device **405** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

(103) The receiver **410** may receive information such as packets, user data, or control information associated with various information channels (e.g., control channels, data channels, and information related to aggregated uplink shared channel transmission for two-step random access channel procedure, or the like). Information may be passed on to other components of the device **405**. The receiver **410** may be an example of aspects of the transceiver **720** described with reference to FIG. **7**. The receiver **410** may utilize a single antenna or a set of antennas.

(104) The UE communications manager **415** may identify a random access message of a random access procedure, the random access message including a random access preamble and a random access payload. The UE communications manager **415** may aggregate multiple PUSCH units of a set of PUSCH resource units associated with one or more POs based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload. The UE communications manager **415** may transmit the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs. One example advantage of aggregating multiple PUSCH units of a set of PUSCH resource units associated with one or more POs based on

one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload may include lower latency relating to communications in random access procedures, as well as extending random access messaging capability and reliability. The UE communications manager **415** may be an example of aspects of the UE communications manager **710** described herein.

(105) The UE communications manager **415**, or its sub-components, may be implemented in hardware, code (e.g., software or firmware) executed by a processor, or any combination thereof. If implemented in code executed by a processor, the functions of the UE communications manager **415**, or its sub-components may be executed by a general-purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described in the present disclosure.

(106) The UE communications manager **415**, or its sub-components, may be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations by one or more physical components. In some examples, the UE communications manager **415**, or its sub-components, may be a separate and distinct component in accordance with various aspects of the present disclosure. In some examples, the UE communications manager **415**, or its sub-components, may be combined with one or more other hardware components, including but not limited to an input/output (I/O) component, a transceiver, a network server, another computing device, one or more other components described in the present disclosure, or a combination thereof in accordance with various aspects of the present disclosure.

(107) The transmitter **420** may transmit signals generated by other components of the device **405**. In some examples, the transmitter **420** may be collocated with a receiver **410** in a transceiver module. For example, the transmitter **420** may be an example of aspects of the transceiver **720** described with reference to FIG. **7**. The transmitter **420** may utilize a single antenna or a set of antennas.

(108) FIG. **5** shows a block diagram **500** of a device **505** that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure. The device **505** may be an example of aspects of a device **405**, or a UE **115** as described herein. The device **505** may include a receiver **510**, a UE communications manager **515**, and a transmitter **535**. The device **505** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

(109) The receiver **510** may receive information such as packets, user data, or control information associated with various information channels (e.g., control channels, data channels, and information related to aggregated uplink shared channel transmission for two-step random access channel procedure, or the like). Information may be passed on to other components of the device **505**. The receiver **510** may be an example of aspects of the transceiver **720** described with reference to FIG. **7**. The receiver **510** may utilize a single antenna or a set of antennas.

(110) The UE communications manager **515** may be an example of aspects of the UE communications manager **415** as described herein. The UE communications manager **515** may include a message component **520**, an aggregation component **525**, and a communications component **530**. The UE communications manager **515** may be an example of aspects of the UE communications manager **710** described herein. The message component **520** may identify a random access message of a random access procedure, the random access message including a random access preamble and a random access payload. The aggregation component **525** may aggregate multiple PUSCH resource units of a set of PUSCH resource units associated with one or more POs based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload. The communications component

530 may transmit the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs. Based on aggregating multiple PUSCH resource units of a set of PUSCH resource units associated with one or more POs, a processor of a UE **115** (e.g., controlling the receiver **710**, the transmitter **740**, or the transceiver **1120** as described with reference to FIG. **11**) may efficiently transmit the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs. The processor of the UE **115** may turn on one or more processing units for transmitting the random access payload of the random access message, increase a processing clock, or a similar mechanism within the UE **115**. As such, when the random access payload of the random access message is transmitted, the processor may be ready to respond more efficiently through the reduction of a ramp up in processing power.

(111) The transmitter **535** may transmit signals generated by other components of the device **505**. In some examples, the transmitter **535** may be collocated with a receiver **510** in a transceiver module. For example, the transmitter **535** may be an example of aspects of the transceiver **720** described with reference to FIG. **7**. The transmitter **535** may utilize a single antenna or a set of antennas.

(112) FIG. **6** shows a block diagram **600** of a UE communications manager **605** that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure. The UE communications manager **605** may be an example of aspects of a UE communications manager **415**, a UE communications manager **515**, or a UE communications manager **710** described herein. The UE communications manager **605** may include a message component **610**, an aggregation component **615**, a communications component **620**, a configuration component **625**, a mapping component **630**, a scheme component **635**, a comparison component **640**, a padding component **645**, and a fallback component **650**. Each of these modules may communicate, directly or indirectly, with one another (e.g., via one or more buses).

(113) The message component **610** may identify a random access message of a random access procedure, the random access message including a random access preamble and a random access payload. In some cases, the random access procedure includes a twostep random access procedure. In some examples, receiving a random access response message may include receiving an indication to perform a random access fallback procedure, where the random access fallback procedure includes a four-step random access procedure. In some examples, the message component **610** may receive a random access response message including an indication to retransmit the random access message.

(114) The aggregation component **615** may aggregate multiple PUSCH resource units of a set of PUSCH resource units associated with one or more POs based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload. In some cases, one or more PUSCH resource units of the set of PUSCH resource units have a default payload size. In some cases, the default payload size includes one or more bits. In some cases, a PUSCH resource unit of the set of PUSCH resource units spans a PO. In some cases, the aggregated multiple PUSCH resource units span over multiple POs.

(115) The communications component **620** may transmit the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs. In some examples, the communications component **620** may transmit the random access payload of the random access message on time and frequency resources of the aggregated multiple PUSCH resource units using a same modulation and coding scheme for each PUSCH resource unit of the aggregated multiple PUSCH resource units. In some examples, the communications component **620** may transmit the random access payload of the random access message on a PUSCH using time and frequency resources

using time and frequency resources that are consecutive or non-consecutive in one or more of a time domain or a frequency domain. In some examples, the communications component **620** may retransmit the random access message based on a random access response message.
(116) The configuration component **625** may receive, from a base station, an indication of an aggregation configuration for the PUSCH resource units. In some examples, aggregating the multiple PUSCH resource units of the set of PUSCH resource units may be further based on the aggregation configuration. In some examples, the configuration component **625** may receive RRC signaling. In some examples, the configuration component **625** may receive one or more of a RAR window for a random access response associated with the random access procedure or a retransmission period of the random access message.
(117) In some examples, the configuration component **625** may receive, from a base station, an indication of one or more of an RO configuration or a PO configuration. In some cases, the RO configuration may include one or more of a periodicity of an RO, a time and frequency resource allocation of the RO, a preamble sequence configuration for a CBRA procedure or a CFRA procedure, an association rule between ROs and POs, or a beam association rule between an SSB or a CSI-RS and the RO. In some cases, the PO configuration may include one or more of a periodicity of a PO, a time and frequency resource allocation of the PO or a PUSCH resource unit, a DMRS configuration, a PUSCH waveform configuration, beam management information for the PUSCH.
(118) In some examples, the configuration component **625** may receive, from a base station, an indication of a power configuration, where transmitting the random access payload of the random access message on the PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs includes transmitting the random access payload of the random access message on time and frequency resources of the aggregated multiple PUSCH resource units using a same transmit power for each PUSCH resource unit of the aggregated multiple PUSCH resource units.
(119) The mapping component **630** may map one or more ROs to the one or more POs based on the one or more of RO configuration or the PO configuration. In some examples, the mapping component **630** may map the random access preamble to the one or more POs in one or more of the time domain or the frequency domain based on the RO configuration or the PO configuration. In some examples, the mapping component **630** may assign, to the one or more POs, available random access preambles including the random access preamble of the random access message associated with one or more ROs. In some examples, the mapping component **630** may map the assigned random access preambles to the one or more POs. In some examples, the mapping component **630** may map the random access payload to one or more of the one or more ROs or the one or more POs in one or more of the time domain or the frequency domain based on a hopping sequence.
(120) In some examples, the mapping component **630** may map a set of DMRS to random access preambles of PUSCH resource units of the aggregated multiple PUSCH resource units based on a mapping rule. In some cases, one or more of the one or more ROs or the one or more POs are contiguous in one or more of the time domain or the frequency domain. In some cases, one or more of the one or more ROs or the one or more POs are noncontiguous in one or more of the time domain or the frequency domain. In some cases, each PUSCH resource unit of the aggregated multiple PUSCH resource units are associated with a unique DMRS of the set of DMRSs. In some cases, the aggregated multiple PUSCH resource units share a same DMRS of the set of DMRS.
(121) The scheme component **635** may determine a modulation and coding scheme capability for the random access message including the random access preamble and the random access payload. In some examples, aggregating the multiple PUSCH resource units of the set of PUSCH resource units associated with the one or more POs is further based on the modulation and coding scheme capability. In some examples, the scheme component **635** may transmit, in one or more of the

random access preamble or a control portion of the random access payload, an indication of the time and frequency resources of the aggregated multiple PUSCH resource units based on the modulation and coding scheme capability supporting the multiple modulation and coding schemes. In some cases, a payload size of one or more PUSCH resource units of the set of PUSCH resource units is fixed based on the modulation and coding scheme capability supporting a single modulation and coding scheme. In some cases, a payload size of one or more PUSCH resource units of the set of PUSCH resource units is variable based on the modulation and coding scheme capability supporting multiple modulation and coding schemes.
(122) The comparison component **640** may compare the payload size of the random access payload to an aggregated payload size of the aggregated multiple PUSCH resource units. In some examples, the comparison component **640** may determine a quantity of unused bits of the aggregated payload size of the aggregated multiple PUSCH resource units based on the comparing. The padding component **645** may pad each unused bit of the quantity of unused bits of the aggregated payload size of the aggregated multiple PUSCH resource units with a null bit. The fallback component **650** may perform the random access fallback procedure based on receiving the random access response message.
(123) FIG. **7** shows a diagram of a system **700** including a device **705** that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure. The device **705** may be an example of or include the components of device **405**, device **505**, or a UE **115** as described herein. The device **705** may include components for bi-directional voice and data communications including components for transmitting and receiving communications, including a UE communications manager **710**, an I/O controller **715**, a transceiver **720**, an antenna **725**, memory **730**, and a processor **740**. These components may be in electronic communication via one or more buses (e.g., bus **745**).
(124) The UE communications manager **710** may identify a random access message of a random access procedure, the random access message including a random access preamble and a random access payload. The UE communications manager **710** may aggregate multiple PUSCH resource units of a set of PUSCH resource units associated with one or more POs based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload. The UE communications manager **710** may transmit the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs.
(125) The I/O controller **715** may manage input and output signals for the device **705**. The I/O controller **715** may also manage peripherals not integrated into the device **705**. In some cases, the I/O controller **715** may represent a physical connection or port to an external peripheral. In some cases, the I/O controller **715** may utilize an operating system such as iOS, ANDROID, MS-DOS, MS-WINDOWS, OS/2, UNIX, LINUX, or another known operating system. In other cases, the I/O controller **715** may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some cases, the I/O controller **715** may be implemented as part of a processor. In some cases, a user may interact with the device **705** via the I/O controller **715** or via hardware components controlled by the I/O controller **715**.
(126) The transceiver **720** may communicate bi-directionally, via one or more antennas, wired, or wireless links as described herein. For example, the transceiver **720** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **720** may also include a modem to modulate the packets and provide the modulated packets to the antennas for transmission, and to demodulate packets received from the antennas. In some cases, the device **705** may include a single antenna **725**. However, in some cases, the device **705** may have more than one antenna **725**, which may be capable of concurrently transmitting or receiving multiple wireless transmissions.
(127) The memory **730** may include random-access memory (RAM) and read-only memory

(ROM). The memory **730** may store computer-readable, computer-executable code **735** including instructions that, when executed, cause the processor to perform various functions described herein. In some cases, the memory **730** may contain, among other things, a basic input/basic output system (BIOS) which may control basic hardware or software operation such as the interaction with peripheral components or devices.

(128) The code **735** may include instructions to implement aspects of the present disclosure, including instructions to support wireless communications. The code **735** may be stored in a non-transitory computer-readable medium such as system memory or other type of memory. In some cases, the code **735** may not be directly executable by the processor **740** but may cause a computer (e.g., when compiled and executed) to perform functions described herein.

(129) The processor **740** may include an intelligent hardware device, (e.g., a general-purpose processor, a DSP, a CPU, a microcontroller, an ASIC, an FPGA, a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). In some cases, the processor **740** may be configured to operate a memory array using a memory controller. In other cases, a memory controller may be integrated into the processor **740**. The processor **740** may be configured to execute computer-readable instructions stored in a memory (e.g., the memory **730**) to cause the device **705** to perform various functions (e.g., functions or tasks supporting aggregated uplink shared channel transmission for two-step random access channel procedure).

(130) FIG. **8** shows a block diagram **800** of a device **805** that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure. The device **805** may be an example of aspects of a base station **105** as described herein. The device **805** may include a receiver **810**, a base station communications manager **815**, and a transmitter **820**. The device **805** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

(131) The receiver **810** may receive information such as packets, user data, or control information associated with various information channels (e.g., control channels, data channels, and information related to aggregated uplink shared channel transmission for two-step random access channel procedure, or the like). Information may be passed on to other components of the device **805**. The receiver **810** may be an example of aspects of the transceiver **1120** described with reference to FIG. **11**. The receiver **810** may utilize a single antenna or a set of antennas.

(132) The base station communications manager **815** may assign a set of PUSCH resource units associated with one or more POs for transmission of a random access payload of a random access message associated with a random access procedure. The base station communications manager **815** may determine an aggregation configuration for multiple PUSCH resource units of the set of PUSCH resource units based on assigning the set of PUSCH resource units. The base station communications manager **815** may transmit, to a UE, an indication of the aggregation configuration for the multiple PUSCH resource units. The base station communications manager **815** may be an example of aspects of the base station communications manager **1110** described herein.

(133) The base station communications manager **815**, or its sub-components, may be implemented in hardware, code (e.g., software or firmware) executed by a processor, or any combination thereof. If implemented in code executed by a processor, the functions of the base station communications manager **815**, or its sub-components may be executed by a general-purpose processor, a DSP, an application-specific integrated circuit (ASIC), an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described in the present disclosure.

(134) The base station communications manager **815**, or its sub-components, may be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations by one or more physical components. In some

examples, the base station communications manager **815**, or its sub-components, may be a separate and distinct component in accordance with various aspects of the present disclosure. In some examples, the base station communications manager **815**, or its sub-components, may be combined with one or more other hardware components, including but not limited to an input/output (I/O) component, a transceiver, a network server, another computing device, one or more other components described in the present disclosure, or a combination thereof in accordance with various aspects of the present disclosure.
(135) The transmitter **820** may transmit signals generated by other components of the device **805**. In some examples, the transmitter **820** may be collocated with a receiver **810** in a transceiver module. For example, the transmitter **820** may be an example of aspects of the transceiver **1120** described with reference to FIG. **11**. The transmitter **820** may utilize a single antenna or a set of antennas.
(136) FIG. **9** shows a block diagram **900** of a device **905** that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure. The device **905** may be an example of aspects of a device **805**, or a base station **105** as described herein. The device **905** may include a receiver **910**, a base station communications manager **915**, and a transmitter **935**. The device **905** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).
(137) The receiver **910** may receive information such as packets, user data, or control information associated with various information channels (e.g., control channels, data channels, and information related to aggregated uplink shared channel transmission for two-step random access channel procedure, or the like). Information may be passed on to other components of the device **905**. The receiver **910** may be an example of aspects of the transceiver **1120** described with reference to FIG. **11**. The receiver **910** may utilize a single antenna or a set of antennas.
(138) The base station communications manager **915** may be an example of aspects of the base station communications manager **815** as described herein. The base station communications manager **915** may include an allocation component **920**, a configuration component **925**, and a communications component **930**. The base station communications manager **915** may be an example of aspects of the base station communications manager **1110** described herein. The allocation component **920** may assign a set of PUSCH resource units associated with one or more POs for transmission of a random access payload of a random access message associated with a random access procedure. The configuration component **925** may determine an aggregation configuration for multiple PUSCH resource units of the set of PUSCH resource units based on assigning the set of PUSCH resource units. The communications component **930** may transmit, to a UE, an indication of the aggregation configuration for the multiple PUSCH resource units.
(139) The transmitter **935** may transmit signals generated by other components of the device **905**. In some examples, the transmitter **935** may be collocated with a receiver **910** in a transceiver module. For example, the transmitter **935** may be an example of aspects of the transceiver **1120** described with reference to FIG. **11**. The transmitter **935** may utilize a single antenna or a set of antennas.
(140) FIG. **10** shows a block diagram **1000** of a base station communications manager **1005** that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure. The base station communications manager **1005** may be an example of aspects of a base station communications manager **815**, a base station communications manager **915**, or a base station communications manager **1110** described herein. The base station communications manager **1005** may include an allocation component **1010**, a configuration component **1015**, a communications component **1020**, a scheme component **1025**, a mapping component **1030**, and a fallback component **1035**. Each of these modules may communicate, directly or indirectly, with one another (e.g., via one or more buses).

(141) The allocation component **1010** may assign a set of PUSCH resource units associated with one or more POs for transmission of a random access payload of a random access message associated with a random access procedure. In some cases, one or more PUSCH resource units of the set of PUSCH resource units may have a default payload size. In some cases, the default payload size includes one or more bits. In some cases, the random access procedure includes a two-step random access procedure. In some cases, a PUSCH resource unit of the set of PUSCH resource units spans a PO. In some cases, the aggregated multiple PUSCH resource units span over multiple POs.
(142) The configuration component **1015** may determine an aggregation configuration for multiple PUSCH resource units of the set of PUSCH resource units based on assigning the set of PUSCH resource units. In some examples, the configuration component **1015** may transmit, to the UE, a second indication of one or more of an RO configuration or a PO configuration. In some examples, configuration component **1015** may transmit, to the UE, a second indication of a power configuration, where receiving the random access payload of the random access message on the PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs includes receiving the random access payload of the random access message on time and frequency resources of the aggregated multiple PUSCH resource units using a same power configuration for each PUSCH resource unit of the aggregated multiple PUSCH resource units. In some cases, the RO configuration may include one or more of: a periodicity of an RO, a time and frequency resource allocation of the RO, a preamble sequence configuration for a CBRA procedure or a CFRA procedure, an association rule between ROs and POs, or a beam association rule between an SSB or a CSI-RS and the RO. In some cases, the PO configuration may include one or more of: a periodicity of a PO, a time and frequency resource allocation of the PO or a PUSCH resource unit, a DMRS configuration, a PUSCH waveform configuration, beam management information for a PUSCH.
(143) The communications component **1020** may transmit, to a UE, an indication of the aggregation configuration for the multiple PUSCH resource units. In some examples, the communications component **1020** may transmit RRC signaling. In some examples, the communications component **1020** may transmit one or more of a RAR window for a random access response associated with the random access procedure or a retransmission period of the random access message. In some examples, the communications component **1020** may receive, from the UE, the random access payload of the random access message on a PUSCH using time and frequency resources of aggregated multiple PUSCH resource units over one or more POs.
(144) The scheme component **1025** may receive, from the UE, a modulation and coding scheme capability for the random access message including the random access preamble and the random access payload. In some examples, determining the aggregation configuration for the multiple PUSCH resource units of the set of PUSCH resource units may be further based on the modulation and coding scheme capability. The mapping component **1030** may map one or more ROs to the one or more POs based on the one or more of RO configuration or the PO configuration. In some examples, the mapping component **1030** may map the random access preamble to the one or more POs in one or more of the time domain or the frequency domain based on the RO configuration or the PO configuration. In some cases, one or more of the one or more ROs or the one or more POs are noncontiguous in one or more of the time domain or the frequency domain. In some cases, one or more of the one or more ROs or the one or more POs are contiguous in one or more of the time domain or the frequency domain.
(145) The fallback component **1035** may transmit a random access response message including an indication to perform a random access fallback procedure, where the random access fallback procedure includes a four-step random access procedure. In some examples, the fallback component **1035** may perform the random access fallback procedure based on the random access response message. In some examples, the fallback component **1035** may transmit a random access

response message including an indication to the UE to retransmit the random access message.
(146) FIG. **11** shows a diagram of a system **1100** including a device **1105** that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure. The device **1105** may be an example of or include the components of device **805**, device **905**, or a base station **105** as described herein. The device **1105** may include components for bi-directional voice and data communications including components for transmitting and receiving communications, including a base station communications manager **1110**, a network communications manager **1115**, a transceiver **1120**, an antenna **1125**, memory **1130**, a processor **1140**, and an inter-station communications manager **1145**. These components may be in electronic communication via one or more buses (e.g., bus **1150**).
(147) The base station communications manager **1110** may assign a set of PUSCH resource units associated with one or more POs for transmission of a random access payload of a random access message associated with a random access procedure. The base station communications manager **1110** may determine an aggregation configuration for multiple PUSCH resource units of the set of PUSCH resource units based on assigning the set of PUSCH resource units. The base station communications manager **1110** may transmit, to a UE, an indication of the aggregation configuration for the multiple PUSCH resource units.
(148) The network communications manager **1115** may manage communications with the core network (e.g., via one or more wired backhaul links). For example, the network communications manager **1115** may manage the transfer of data communications for client devices, such as one or more UEs **115**.
(149) The transceiver **1120** may communicate bi-directionally, via one or more antennas, wired, or wireless links as described herein. For example, the transceiver **1120** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **1120** may also include a modem to modulate the packets and provide the modulated packets to the antennas for transmission, and to demodulate packets received from the antennas. In some cases, the device **1105** may include a single antenna **1125**. However, in some cases, the device **1105** may have more than one antenna **1125**, which may be capable of concurrently transmitting or receiving multiple wireless transmissions.
(150) The memory **1130** may include RAM, ROM, or a combination thereof. The memory **1130** may store computer-readable code **1135** including instructions that, when executed by a processor (e.g., the processor **1140**) cause the device to perform various functions described herein. In some cases, the memory **1130** may contain, among other things, a BIOS which may control basic hardware or software operation such as the interaction with peripheral components or devices.
(151) The code **1135** may include instructions to implement aspects of the present disclosure, including instructions to support wireless communications. The code **1135** may be stored in a non-transitory computer-readable medium such as system memory or other type of memory. In some cases, the code **1135** may not be directly executable by the processor **1140** but may cause a computer (e.g., when compiled and executed) to perform functions described herein.
(152) The processor **1140** may include an intelligent hardware device, (e.g., a general-purpose processor, a DSP, a CPU, a microcontroller, an ASIC, an FPGA, a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). In some cases, the processor **1140** may be configured to operate a memory array using a memory controller. In some cases, a memory controller may be integrated into processor **1140**. The processor **1140** may be configured to execute computer-readable instructions stored in a memory (e.g., the memory **1130**) to cause the device **1105** to perform various functions (e.g., functions or tasks supporting aggregated uplink shared channel transmission for two-step random access channel procedure).
(153) The inter-station communications manager **1145** may manage communications with other base station **105**, and may include a controller or scheduler for controlling communications with

UEs **115** in cooperation with other base stations **105**. For example, the inter-station communications manager **1145** may coordinate scheduling for transmissions to UEs **115** for various interference mitigation techniques such as beamforming or joint transmission. In some examples, the inter-station communications manager **1145** may provide an X2 interface within an LTE/LTE-A wireless communications network technology to provide communication between base stations **105**.

(154) FIG. **12** shows a flowchart illustrating a method **1200** that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure. The operations of method **1200** may be implemented by a UE **115** or its components as described herein. For example, the operations of method **1200** may be performed by a UE communications manager as described with reference to FIGS. **4** through **7**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the functions described herein. Additionally or alternatively, a UE may perform aspects of the functions described herein using special-purpose hardware.

(155) At **1205**, the UE may identify a random access message of a random access procedure, the random access message including a random access preamble and a random access payload. The operations of **1205** may be performed according to the methods described herein. In some examples, aspects of the operations of **1205** may be performed by a message component as described with reference to FIGS. **4** through **7**.

(156) At **1210**, the UE may aggregate multiple PUSCH resource units of a set of PUSCH resource units associated with one or more POs based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload. The operations of **1210** may be performed according to the methods described herein. In some examples, aspects of the operations of **1210** may be performed by an aggregation component as described with reference to FIGS. **4** through **7**.

(157) At **1215**, the UE may transmit the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs. The operations of **1215** may be performed according to the methods described herein. In some examples, aspects of the operations of **1215** may be performed by a communications component as described with reference to FIGS. **4** through **7**.

(158) FIG. **13** shows a flowchart illustrating a method **1300** that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure. The operations of method **1300** may be implemented by a UE **115** or its components as described herein. For example, the operations of method **1300** may be performed by a UE communications manager as described with reference to FIGS. **4** through **7**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the functions described herein.

(159) Additionally or alternatively, a UE may perform aspects of the functions described herein using special-purpose hardware.

(160) At **1305**, the UE may identify a random access message of a random access procedure, the random access message including a random access preamble and a random access payload. The operations of **1305** may be performed according to the methods described herein. In some examples, aspects of the operations of **1305** may be performed by a message component as described with reference to FIGS. **4** through **7**.

(161) At **1310**, the UE may aggregate multiple PUSCH resource units of a set of PUSCH resource units associated with one or more POs based on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload. The operations of **1310** may be performed according to the methods described herein. In some examples, aspects of the operations of **1310** may be performed by an aggregation component as described with reference to FIGS. **4** through **7**.

(162) At **1315**, the UE may receive, from a base station, an indication of one or more of an RO configuration or a PO configuration. The operations of **1315** may be performed according to the methods described herein. In some examples, aspects of the operations of **1315** may be performed by a configuration component as described with reference to FIGS. **4** through **7**.
(163) At **1320**, the UE may map one or more ROs to the one or more POs based on the one or more of RO configuration or the PO configuration. The operations of **1320** may be performed according to the methods described herein. In some examples, aspects of the operations of **1320** may be performed by a mapping component as described with reference to FIGS. **4** through **7**.
(164) At **1325**, the UE may transmit the random access payload of the random access message on a PUSCH using time and frequency resources of the aggregated multiple PUSCH resource units over the one or more POs. The operations of **1325** may be performed according to the methods described herein. In some examples, aspects of the operations of **1325** may be performed by a communications component as described with reference to FIGS. **4** through **7**.
(165) FIG. **14** shows a flowchart illustrating a method **1400** that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure. The operations of method **1400** may be implemented by a base station **105** or its components as described herein. For example, the operations of method **1400** may be performed by a base station communications manager as described with reference to FIGS. **8** through **11**. In some examples, a base station may execute a set of instructions to control the functional elements of the base station to perform the functions described herein. Additionally or alternatively, a base station may perform aspects of the functions described herein using special-purpose hardware.
(166) At **1405**, the base station may assign a set of PUSCH resource units associated with one or more POs for transmission of a random access payload of a random access message associated with a random access procedure. The operations of **1405** may be performed according to the methods described herein. In some examples, aspects of the operations of **1405** may be performed by an allocation component as described with reference to FIGS. **8** through **11**.
(167) At **1410**, the base station may determine an aggregation configuration for multiple PUSCH resource units of the set of PUSCH resource units based on assigning the set of PUSCH resource units. The operations of **1410** may be performed according to the methods described herein. In some examples, aspects of the operations of **1410** may be performed by a configuration component as described with reference to FIGS. **8** through **11**.
(168) At **1415**, the base station may transmit an indication of the aggregation configuration for the multiple PUSCH resource units. The operations of **1415** may be performed according to the methods described herein. In some examples, aspects of the operations of **1415** may be performed by a communications component as described with reference to FIGS. **8** through **11**.
(169) FIG. **15** shows a flowchart illustrating a method **1500** that supports aggregated uplink shared channel transmission for two-step random access channel procedure in accordance with aspects of the present disclosure. The operations of method **1500** may be implemented by a base station **105** or its components as described herein. For example, the operations of method **1500** may be performed by a base station communications manager as described with reference to FIGS. **8** through **11**. In some examples, a base station may execute a set of instructions to control the functional elements of the base station to perform the functions described herein. Additionally or alternatively, a base station may perform aspects of the functions described herein using special-purpose hardware.
(170) At **1505**, the base station may assign a set of PUSCH resource units associated with one or more POs for transmission of a random access payload of a random access message associated with a random access procedure. The operations of **1505** may be performed according to the methods described herein. In some examples, aspects of the operations of **1505** may be performed by an allocation component as described with reference to FIGS. **8** through **11**.

(171) At **1510**, the base station may determine an aggregation configuration for multiple PUSCH resource units of the set of PUSCH resource units based on assigning the set of PUSCH resource units. The operations of **1510** may be performed according to the methods described herein. In some examples, aspects of the operations of **1510** may be performed by a configuration component as described with reference to FIGS. **8** through **11**.
(172) At **1515**, the base station may transmit an indication of one or more of an RO configuration or a PO configuration. The operations of **1515** may be performed according to the methods described herein. In some examples, aspects of the operations of **1515** may be performed by a configuration component as described with reference to FIGS. **8** through **11**.
(173) At **1520**, the base station may map one or more ROs to the one or more POs based on the one or more of RO configuration or the PO configuration. The operations of **1520** may be performed according to the methods described herein. In some examples, aspects of the operations of **1520** may be performed by a mapping component as described with reference to FIGS. **8** through **11**.
(174) At **1525**, the base station may transmit a second indication of the aggregation configuration for the multiple PUSCH resource units. The operations of **1525** may be performed according to the methods described herein. In some examples, aspects of the operations of **1525** may be performed by a communications component as described with reference to FIGS. **8** through **11**.
(175) It should be noted that the methods described herein describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Further, aspects from two or more of the methods may be combined.
(176) Techniques described herein may be used for various wireless communications systems such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal frequency division multiple access (OFDMA), single carrier frequency division multiple access (SC-FDMA), and other systems. A CDMA system may implement a radio technology such as CDMA**2000**, Universal Terrestrial Radio Access (UTRA), etc. CDMA2000 covers IS-2000, IS-95, and IS-856 standards. IS-2000 Releases may be commonly referred to as CDMA2000 1X, 1X, etc. IS-856 (TIA-856) is commonly referred to as CDMA2000 1xEV-DO, High Rate Packet Data (HRPD), etc. UTRA includes Wideband CDMA (WCDMA) and other variants of CDMA. A TDMA system may implement a radio technology such as Global System for Mobile Communications (GSM).
(177) An OFDMA system may implement a radio technology such as Ultra Mobile Broadband (UMB), Evolved UTRA (E-UTRA), Institute of Electrical and Electronics Engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Flash-OFDM, etc. UTRA and E-UTRA are part of Universal Mobile Telecommunications System (UMTS). LTE, LTE-A, and LTE-A Pro are releases of UMTS that use E-UTRA. UTRA, E-UTRA, UMTS, LTE, LTE-A, LTE-A Pro, NR, and GSM are described in documents from the organization named “3rd Generation Partnership Project” (3GPP). CDMA2000 and UMB are described in documents from an organization named “3rd Generation Partnership Project 2” (3GPP2). The techniques described herein may be used for the systems and radio technologies mentioned herein as well as other systems and radio technologies. While aspects of an LTE, LTE-A, LTE-A Pro, or NR system may be described for purposes of example, and LTE, LTE-A, LTE-A Pro, or NR terminology may be used in much of the description, the techniques described herein are applicable beyond LTE, LTE-A, LTE-A Pro, or NR applications.
(178) A macro cell generally covers a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by UEs with service subscriptions with the network provider. A small cell may be associated with a lower-powered base station, as compared with a macro cell, and a small cell may operate in the same or different (e.g., licensed, unlicensed, etc.) frequency bands as macro cells. Small cells may include pico cells, femto cells, and micro cells according to various examples. A pico cell, for example, may cover a small geographic area and may allow unrestricted access by UEs with service subscriptions with the network provider. A

femto cell may also cover a small geographic area (e.g., a home) and may provide restricted access by UEs having an association with the femto cell (e.g., UEs in a closed subscriber group (CSG), UEs for users in the home, and the like). An eNB for a macro cell may be referred to as a macro eNB. An eNB for a small cell may be referred to as a small cell eNB, a pico eNB, a femto eNB, or a home eNB. An eNB may support one or multiple (e.g., two, three, four, and the like) cells, and may also support communications using one or multiple component carriers.

(179) The wireless communications systems described herein may support synchronous or asynchronous operation. For synchronous operation, the base stations may have similar frame timing, and transmissions from different base stations may be approximately aligned in time. For asynchronous operation, the base stations may have different frame timing, and transmissions from different base stations may not be aligned in time. The techniques described herein may be used for either synchronous or asynchronous operations.

(180) Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

(181) The various illustrative blocks and modules described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, an FPGA, or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

(182) The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described herein can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

(183) Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, non-transitory computer-readable media may include RAM, ROM, electrically erasable programmable ROM (EEPROM), flash memory, compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, include CD, laser disc, optical

disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of computer-readable media.

(184) As used herein, including in the claims, “or” as used in a list of items (e.g., a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

(185) In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label, or other subsequent reference label.

(186) The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “exemplary” used herein means “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.

(187) The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein, but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. A method for wireless communication at a user equipment (UE), comprising: identifying a random access message of a random access procedure, the random access message comprising a random access preamble and a random access payload; aggregating, by the UE, multiple physical uplink shared channel resource units of a set of physical uplink shared channel resource units associated with multiple physical uplink shared channel occasions based at least in part on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload; determining a modulation and coding scheme capability of the UE for the random access message comprising the random access preamble and the random access payload, wherein aggregating the multiple physical uplink shared channel resource units of the set of physical uplink shared channel resource units associated with the one or more physical uplink shared channel occasions is further based at least in part on the modulation and coding scheme capability, wherein a payload size of one or more physical uplink shared channel resource units of the set of physical uplink shared channel resource units is: (i) fixed based at least in part on the modulation and coding scheme capability supporting a single modulation and coding scheme; or (ii) variable based at least in part on the modulation and coding scheme capability supporting multiple modulation and coding schemes; and transmitting the random access

payload of the random access message on a physical uplink shared channel using time and frequency resources of the aggregated multiple physical uplink shared channel resource units over the multiple physical uplink shared channel occasions.

2. The method of claim 1, further comprising: receiving, from a base station, an indication of an aggregation configuration for the multiple physical uplink shared channel resource units, wherein aggregating the multiple physical uplink shared channel resource units of the set of physical uplink shared channel resource units is further based at least in part on the aggregation configuration.

3. The method of claim 2, wherein receiving the indication of the aggregation configuration for the multiple physical uplink shared channel resource units comprises: receiving radio resource control signaling.

4. The method of claim 2, wherein receiving the indication comprises: receiving one or more of a random access response window for a random access response associated with the random access procedure or a retransmission period of the random access message.

5. The method of claim 2, wherein the aggregation configuration for the multiple PUSCH resource units of the set of PUSCH resource units is based on the modulation and coding scheme capability.

6. The method of claim 1, further comprising: receiving, from a base station, an indication of one or more of a random access channel occasion configuration or a physical uplink shared channel occasion configuration; and mapping one or more random access channel occasions to the multiple physical uplink shared channel occasions based at least in part on the one or more of the random access channel occasion configuration or the physical uplink shared channel occasion configuration.

7. The method of claim 6, wherein one or more of the one or more random access channel occasions or the multiple physical uplink shared channel occasions are contiguous in one or more of a time domain or a frequency domain.

8. The method of claim 6, wherein one or more of the one or more random access channel occasions or the multiple physical uplink shared channel occasions are noncontiguous in one or more of a time domain or a frequency domain.

9. The method of claim 6, wherein mapping the one or more random access channel occasions to the multiple physical uplink shared channel occasions comprises: mapping the random access preamble to the multiple physical uplink shared channel occasions in one or more of a time domain or a frequency domain based at least in part on the random access channel occasion configuration or the physical uplink shared channel occasion configuration.

10. The method of claim 6, wherein mapping the one or more random access channel occasions to the multiple physical uplink shared channel occasions comprises: assigning available random access preambles including the random access preamble of the random access message associated with the one or more random access channel occasions to the multiple physical uplink shared channel occasions; and mapping the assigned random access preambles to the multiple physical uplink shared channel occasions.

11. The method of claim 6, wherein the random access channel occasion configuration comprises one or more of: a periodicity of a random access channel occasion, a time and frequency resource allocation of the random access channel occasion, a preamble sequence configuration for a contention-based random access procedure or a contention-free random access procedure, an association rule between random access channel occasions and physical uplink shared channel occasions, or a beam association rule between a synchronization signal blocks or a channel state information reference signal and the random access channel occasion.

12. The method of claim 6, wherein the physical uplink shared channel occasion configuration comprises one or more of: a periodicity of a physical uplink shared channel occasion, a time and frequency resource allocation of the physical uplink shared channel occasion or a physical uplink shared channel resource unit, a demodulation reference signal configuration, a physical uplink shared channel waveform configuration, beam management information for the physical uplink

shared channel.
13. The method of claim 1, further comprising: mapping the random access payload to one or more of one or more random access channel occasions or the multiple physical uplink shared channel occasions in one or more of a time domain or a frequency domain based at least in part on a hopping sequence.
14. The method of claim 1, wherein transmitting the random access payload of the random access message on the physical uplink shared channel using the time and frequency resources of the aggregated multiple physical uplink shared channel resource units over the multiple physical uplink shared channel occasions comprises: transmitting the random access payload of the random access message on the time and frequency resources of the aggregated multiple physical uplink shared channel resource units using a same modulation and coding scheme for each physical uplink shared channel resource unit of the aggregated multiple physical uplink shared channel resource units.
15. The method of claim 1, wherein transmitting the random access payload of the random access message on the physical uplink shared channel using the time and frequency resources of the aggregated multiple physical uplink shared channel resource units over the multiple physical uplink shared channel occasions comprises: transmitting, in one or more of the random access preamble or a control portion of the random access payload, an indication of the time and frequency resources of the aggregated multiple physical uplink shared channel resource units based at least in part on the modulation and coding scheme capability supporting multiple modulation and coding schemes.
16. The method of claim 1, further comprising: mapping a set of demodulation reference signals to random access preambles of physical uplink shared channel resource units of the aggregated multiple physical uplink shared channel resource units based at least in part on a mapping rule.
17. The method of claim 16, wherein each physical uplink shared channel resource unit of the aggregated multiple physical uplink shared channel resource units are associated with a unique demodulation reference signal of the set of demodulation reference signals.
18. The method of claim 16, wherein the aggregated multiple physical uplink shared channel resource units share a same demodulation reference signal of the set of demodulation reference signals.
19. The method of claim 1, wherein aggregating the multiple physical uplink shared channel resource units of the set of physical uplink shared channel resource units associated with the multiple physical uplink shared channel occasions comprises: comparing the payload size of the random access payload to an aggregated payload size of the aggregated multiple physical uplink shared channel resource units; determining a quantity of unused bits of the aggregated payload size of the aggregated multiple physical uplink shared channel resource units based at least in part on the comparing; and padding each unused bit of the quantity of unused bits of the aggregated payload size of the aggregated multiple physical uplink shared channel resource units with a null bit.
20. The method of claim 1, further comprising: receiving, from a base station, an indication of a power configuration, wherein transmitting the random access payload of the random access message on the physical uplink shared channel using the time and frequency resources of the aggregated multiple physical uplink shared channel resource units over the multiple physical uplink shared channel occasions comprises: transmitting the random access payload of the random access message on the time and frequency resources of the aggregated multiple physical uplink shared channel resource units using a same transmit power for each physical uplink shared channel resource unit of the aggregated multiple physical uplink shared channel resource units.
21. The method of claim 1, wherein one or more physical uplink shared channel resource units of the set of physical uplink shared channel resource units have a default payload size.
22. The method of claim 1, wherein transmitting the random access payload of the random access message on the physical uplink shared channel using the time and frequency resources comprises: transmitting using the time and frequency resources that are consecutive in one or more of a time

domain or a frequency domain.

23. The method of claim 1, further comprising: receiving a random access response message comprising an indication to perform a random access fallback procedure, wherein the random access fallback procedure comprises a four-step random access procedure; and performing the random access fallback procedure based at least in part on receiving the random access response message.

24. The method of claim 1, further comprising: receiving a random access response message comprising an indication to retransmit the random access message; and retransmitting the random access message based at least in part on the random access response message.

25. The method of claim 1, wherein a physical uplink shared channel resource unit of the set of physical uplink shared channel resource units spans a physical uplink shared channel occasion; and wherein the aggregated multiple physical uplink shared channel resource units span over the multiple physical uplink shared channel occasions.

26. A user equipment (UE) for wireless communication, comprising: one or more processors, one or more memories coupled with the one or more processors; and instructions stored in the one or more memories and executable by the one or more processors to cause the UE to: identify a random access message of a random access procedure, the random access message comprising a random access preamble and a random access payload; aggregate, by the UE, multiple physical uplink shared channel resource units of a set of physical uplink shared channel resource units associated with multiple physical uplink shared channel occasions based at least in part on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload; determine a modulation and coding scheme capability of the UE for the random access message comprising the random access preamble and the random access payload, wherein aggregating the multiple physical uplink shared channel resource units of the set of physical uplink shared channel resource units associated with the one or more physical uplink shared channel occasions is further based at least in part on the modulation and coding scheme capability, wherein a payload size of one or more physical uplink shared channel resource units of the set of physical uplink shared channel resource units is (i) fixed based at least in part on the modulation and coding scheme capability supporting a single modulation and coding scheme; or (ii) variable based at least in part on the modulation and coding scheme capability supporting multiple modulation and coding schemes; and transmit the random access payload of the random access message on a physical uplink shared channel using time and frequency resources of the aggregated multiple physical uplink shared channel resource units over the multiple physical uplink shared channel occasions.

27. A user equipment (UE) for wireless communication, comprising: means for identifying a random access message of a random access procedure, the random access message comprising a random access preamble and a random access payload; means for aggregating, by the UE, multiple physical uplink shared channel resource units of a set of physical uplink shared channel resource units associated with multiple physical uplink shared channel occasions based at least in part on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload; means for determining a modulation and coding scheme capability of the UE for the random access message comprising the random access preamble and the random access payload, wherein aggregating the multiple physical uplink shared channel resource units of the set of physical uplink shared channel resource units associated with the one or more physical uplink shared channel occasions is further based at least in part on the modulation and coding scheme capability, wherein a payload size of one or more physical uplink shared channel resource units of the set of physical uplink shared channel resource units is (i) fixed based at least in part on the modulation and coding scheme capability supporting a single modulation and coding scheme; or (ii) variable based at least in part on the modulation and coding scheme capability supporting multiple modulation and coding schemes; and means for transmitting

the random access payload of the random access message on a physical uplink shared channel using time and frequency resources of the aggregated multiple physical uplink shared channel resource units over the multiple physical uplink shared channel occasions.

28. A non-transitory computer-readable medium storing code for wireless communication at a user equipment (UE), the code comprising instructions executable by a processor to: identify a random access message of a random access procedure, the random access message comprising a random access preamble and a random access payload; aggregate, by the UE, multiple physical uplink shared channel resource units of a set of physical uplink shared channel resource units associated with multiple physical uplink shared channel occasions based at least in part on one or more of a payload size of the random access payload or a modulation and coding scheme associated with the random access payload; determine a modulation and coding scheme capability of the UE for the random access message comprising the random access preamble and the random access payload, wherein aggregating the multiple physical uplink shared channel resource units of the set of physical uplink shared channel resource units associated with the one or more physical uplink shared channel occasions is further based at least in part on the modulation and coding scheme capability, wherein a payload size of one or more physical uplink shared channel resource units of the set of physical uplink shared channel resource units is (i) fixed based at least in part on the modulation and coding scheme capability supporting a single modulation and coding scheme; or (ii) variable based at least in part on the modulation and coding scheme capability supporting multiple modulation and coding schemes; and transmit the random access payload of the random access message on a physical uplink shared channel using time and frequency resources of the aggregated multiple physical uplink shared channel resource units over the multiple physical uplink shared channel occasions.